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Compositional Difficulty in Multi-Hop Question Answering

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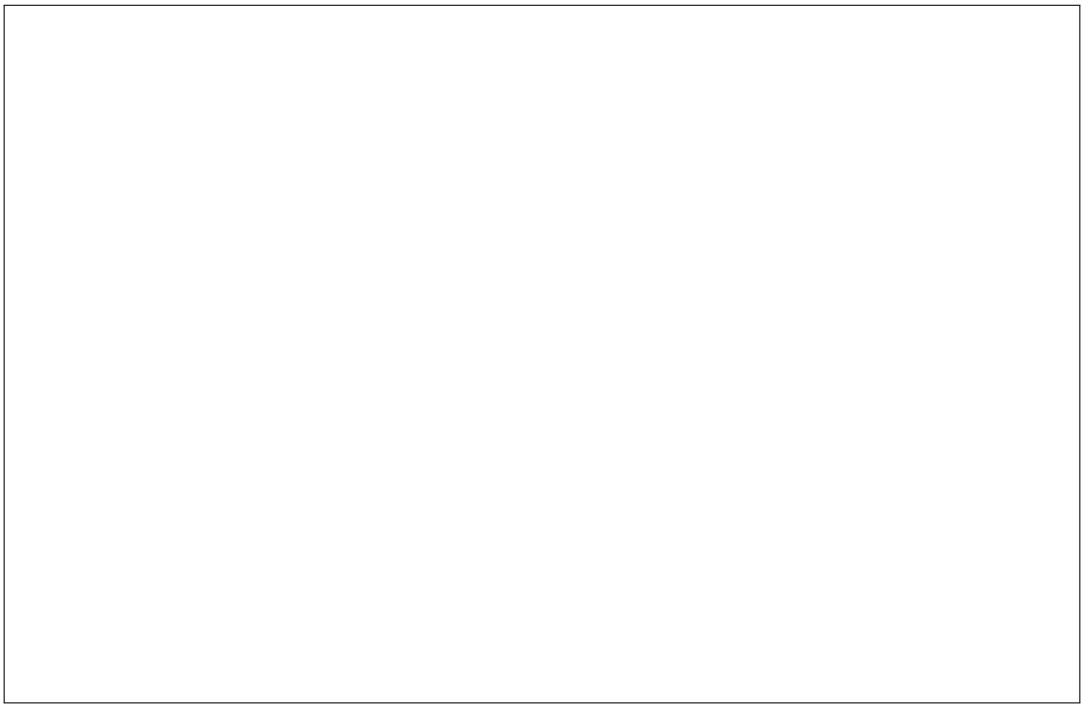

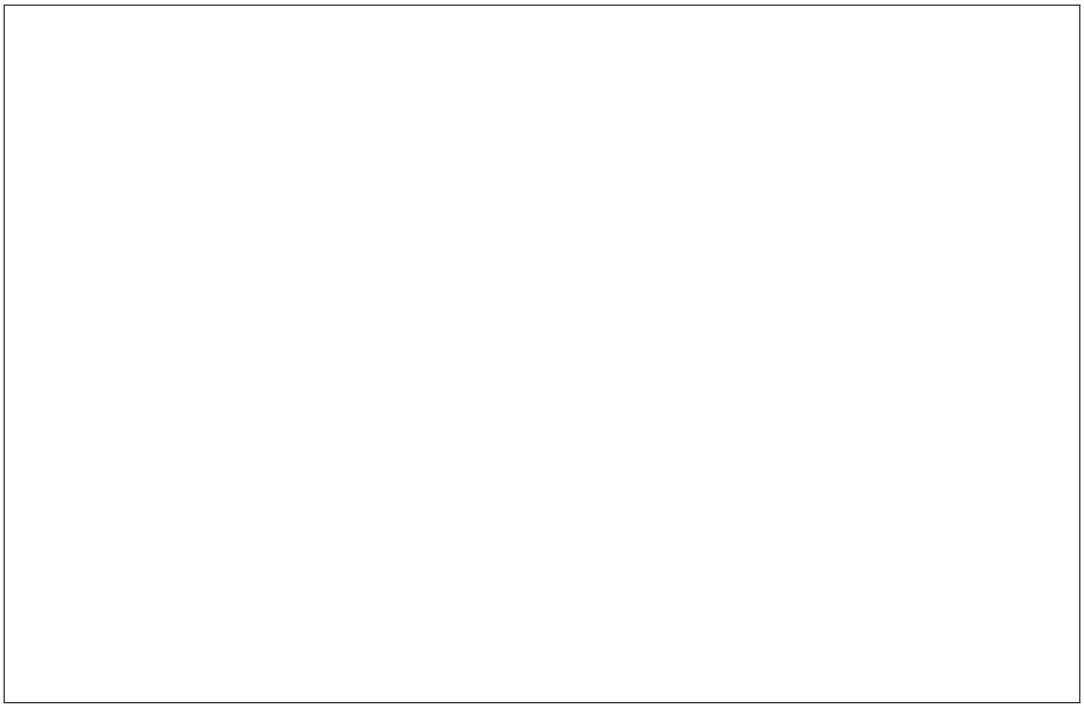

Abstract

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List of Abbreviations

AI	Artificial Intelligence
AI4DM	AI for Decision Making Lab, Concordia University
AUC	Area Under the Receiver Operating Characteristic Curve
BM25	Best Match 25, a probabilistic lexical ranking function
ColBERT	Contextualized Late Interaction over BERT
DAG	Directed Acyclic Graph
DPR	Dense Passage Retrieval
EPT	Tunisia Polytechnic School (*École Polytechnique de Tunisie*)
EV	Engineering, Computer Science and Visual Arts Complex, Concordia University
FLARE	Forward-Looking Active Retrieval Augmented Generation
IRCoT	Interleaving Retrieval with Chain-of-Thought Reasoning
LDA	Linear Discriminant Analysis
LLM	Large Language Model
MDR	Multi-Hop Dense Retrieval
MIAE	Mechanical, Industrial and Aerospace Engineering, Concordia University
RAG	Retrieval-Augmented Generation
ROC	Receiver Operating Characteristic
TF–IDF	Term Frequency–Inverse Document Frequency

General Introduction

Chapter 1
General Presentation and Context of the Internship

Introduction

The purpose of this chapter is to situate the project within its general research and institutional context, and to present the main objectives, research directions, and methodology that guided the work carried out during the internship.

The work reported in this thesis was conducted as a research internship at the AI for Decision Making Lab of Concordia University, under the supervision of Dr. Yassine Yaakoubi, and constitutes the final-year project of the engineering curriculum at École Polytechnique de Tunisie (EPT).

This chapter first introduces the host institution and the research laboratory in which the internship was carried out, together with their main areas of research. It then presents the broader scientific context of the internship, centered on large language models, retrieval-augmented generation, and multi-hop question answering. The progression of the research work is subsequently outlined, from the initial literature exploration and systematic study of trust signals in RAG systems, to the investigation of compositional question difficulty in language model representations, and finally to the development of a retrieval-oriented framework for multi-hop question decomposition. The chapter concludes by presenting the main objectives, contributions, and research methodology adopted throughout the internship.

1.1 Host organisation

The internship was hosted by the **AI for Decision Making Lab (AI4DM)**, a research group within the Department of Mechanical, Industrial and Aerospace Engineering (MIAE) of the Gina Cody School of Engineering and Computer Science at Concordia University, Montreal, Canada. Table 1.1 summarises the host organisation. The following subsections present the university, the faculty and department to which the laboratory belongs, and finally the research group in which the work was carried out.

Table 1.1: Host organisation details

Institution	Concordia University
Faculty	Gina Cody School of Engineering and Computer Science
Department	Mechanical, Industrial and Aerospace Engineering (MIAE)
Laboratory	AI for Decision Making Lab (AI4DM)
Supervisor	Dr. Yassine Yaakoubi
Location	Montreal, Quebec, Canada
Research axes	Optimisation, learning and simulation for large-scale decision making under uncertainty; large language models and retrieval-augmented generation; uncertainty quantification and reliability of AI systems

1.1.1 Concordia University

Concordia University is a public comprehensive research university located in Montreal, Quebec. It was officially founded in 1974 following the merger of Loyola College, established in 1896, and Sir George Williams University, whose origins can be traced to the educational activities of the Montreal YMCA [N1, N2]. The name of the university derives from the motto of the City of Montreal, *Concordia salus*, meaning "well-being through harmony."

Concordia operates two main campuses: the Sir George Williams Campus in downtown Montreal and the Loyola Campus in the west of the city. In the 2024–2025 academic year, 43,922 students were enrolled in credit courses across the university [N3]. Its academic activities cover a broad range of disciplines through the Faculty of Arts and Science, the Gina Cody School of Engineering and Computer Science, the John Molson School of Business, and the Faculty of Fine Arts.

1.1.2 Gina Cody School of Engineering and Computer Science

The Gina Cody School of Engineering and Computer Science is Concordia University's faculty dedicated to engineering and computer science. In 2018, following a fifteen-million-dollar donation from Concordia alumna Gina Parvaneh Cody, the faculty was renamed in her honour, becoming the first engineering faculty in Canada to be named after a woman [N4]. Gina Cody, who earned her doctorate in building engineering from Concordia in 1989 and was the first woman in Canada to obtain a PhD in the discipline, has served as Chancellor of Concordia University since January 2025 [N5].

The school brings together 11,315 undergraduate and graduate students, 262 faculty members, 43 undergraduate and graduate programmes, and 20 interdisciplinary research centres [N6]. Its academic structure comprises seven departments, including the Department of Mechanical, Industrial and Aerospace Engineering (MIAE), to which the host research group belongs.

Teaching and research activities are conducted principally in the Engineering, Computer Science and Visual Arts Integrated Complex (EV Building), located on Concordia's Sir George Williams Campus and shown in Figure 1.1.

Figure 1.1: The EV Complex, home of the Gina Cody School (photograph: Tasitch, CC BY 2.0)

1.1.3 Research group, supervision, and research axes

The work presented in this report was conducted within the AI for Decision Making Lab (AI4DM), led by Dr. Yassine Yaakoubi, Assistant Professor in the Department of Mechanical, Industrial and Aerospace Engineering at Concordia University.

The research activities of the group lie at the intersection of optimisation, machine learning, and simulation for large-scale real-world decision making. The group combines mathematical optimisation methods, including mixed-integer and stochastic programming and decomposition techniques, with machine-learning approaches such as deep learning, reinforcement learning, and graph-based models. The general objective is to design decision-support systems that remain efficient, robust, and scalable in complex and uncertain environments, with applications including logistics, transportation, supply chains, and sustainability.

A major part of the group's work is organised around machine-learning-augmented optimisation, where learned models are integrated into classical optimisation procedures, and end-to-end optimisation learning, where learning mechanisms are used more directly to guide decision making.

A further strand of the group's research activity extends these concerns with robustness, uncertainty, and adaptive decision making to systems built on large language models. This includes retrieval-augmented generation, where language models are grounded in evidence retrieved from external knowledge sources, as well as uncertainty quantification, reliability assessment, and adaptive behaviour in such systems.

The work carried out during this internship belongs primarily to this latter research direction. It focuses on multi-hop question answering in retrieval-augmented systems and investigates, from complementary perspectives, how the difficulty of such questions can be understood and how their processing can be made more effective. The research therefore evolved from the study of reliability and actionable signals in RAG systems, to the analysis of compositional question difficulty in the internal representations of large

language models, and finally to the development of a retrieval-oriented approach to multi-hop question decomposition.

This research environment provided the basis for the successive studies conducted during the internship and for the formulation of the main research problem addressed in this thesis.

1.2 Problem statement

1.3 Objectives and expected deliverables

1.4 Working methodology

Conclusion

Chapter 2
Background and State of the Art

Introduction

Multi-hop question answering requires a system to retrieve and combine several pieces of evidence before producing an answer. In Retrieval-Augmented Generation (RAG), this creates a particular difficulty: the evidence needed for a complex question may not be easily retrieved from the original query, and different questions may require very different amounts of retrieval and reasoning.

This chapter introduces the concepts needed for the remainder of the thesis. It first presents RAG and multi-hop question answering, then reviews retrieval methods and the notions of evidence coverage and retrieval depth. It subsequently discusses adaptive computation, probing of internal model representations, and question decomposition, before focusing on decomposition as a means of making evidence easier to retrieve. The chapter ends by identifying the two research gaps addressed in the following chapters.

2.1 Multi-Hop Question Answering and Retrieval-Augmented Generation

2.1.1 Retrieval-Augmented Generation

Large Language Models (LLMs) acquire substantial knowledge during pre-training, but their parametric knowledge is neither a complete nor a continuously updated record of the world. It may omit specialised facts, become outdated, or be insufficient when a question requires information from a particular source. RAG complements the model with an external corpus that can be searched at inference time [1].

Let the corpus be

$$\mathcal{C} = \{d_1, d_2, \ldots, d_N\}, \tag{2.1}$$

where each d_i is a document or passage. For a query q, a retriever assigns a relevance score $s(q, d)$ and returns

$$D_k(q) = \operatorname*{TopK}_{d \in \mathcal{C}} s(q, d). \tag{2.2}$$

The generator then predicts an answer y conditional on the question and the retrieved passages,

$$p\big(y \mid q, D_k(q)\big). \tag{2.3}$$

This division of labour is central: the retriever determines which external information becomes visible, while the generator interprets and combines that information. Figure 2.1

depicts the separation between the two components. A stronger generator can reason more effectively over retrieved context, but it cannot directly exploit a relevant passage that the retriever never exposes.

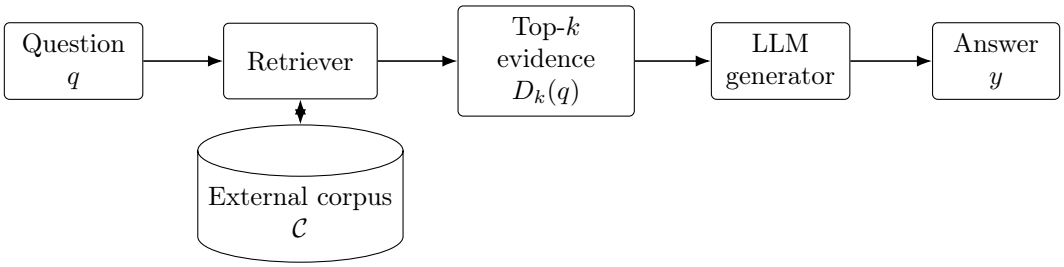


Figure 2.1: A conventional RAG pipeline. Retrieval selects a small portion of the external corpus; only those passages are available to the language model during generation.

The retrieval cutoff k controls both opportunity and cost. A larger context may contain more useful evidence, but also introduces distractors and increases the amount of text processed by the generator. This tension is mild when one passage directly answers the question and more consequential when the answer depends on several passages with different lexical and semantic relations to the original query.

2.1.2 Multi-Hop Question Answering

A single-hop question can normally be resolved through one principal factual relation or evidence source. A multi-hop question instead requires several dependent facts or operations,

$$r_1 \longrightarrow r_2 \longrightarrow \cdots \longrightarrow r_h, \tag{2.4}$$

where an intermediate result may determine what must be resolved next. The entity or value transferred between two steps is called a *bridge entity*. For example, the question *Which country is the birthplace of the author of Book X?* first requires identifying the author, then retrieving that person's birthplace. Figure 2.2 makes this dependency explicit.

Reasoning hop count and retrieval depth describe different quantities and must not be conflated. A four-hop benchmark question encodes four dependent reasoning operations. Retrieval depth $k = 50$, by contrast, means that up to fifty ranked passages are considered for a query. Later chapters seek to reduce the latter; they do not alter the benchmark's compositional hop count.

2.1.3 Multi-Hop Question Answering Benchmarks

Multi-hop benchmarks operationalise compositional reasoning in different ways. Three datasets are particularly relevant to this thesis. **HotpotQA** is a Wikipedia-based benchmark designed for diverse, explainable multi-hop question answering. In addition to answers, it annotates supporting facts, enabling joint evaluation of answer prediction and evidence identification [2].

2WikiMultiHopQA constructs multi-hop questions from Wikipedia and Wikidata information and provides explicit evidence and reasoning structures. Its question types include comparison and compositional chains, allowing models to be assessed at intermediate as well as final steps [3].

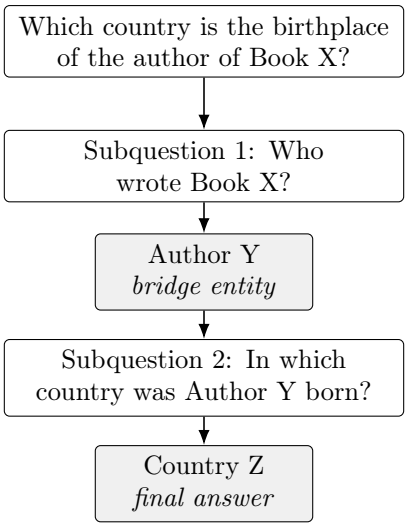


Figure 2.2: A two-hop reasoning chain. The answer to the first subquestion supplies a bridge entity needed to formulate and resolve the second information need.

MuSiQue is constructed bottom-up by composing connected single-hop questions into questions requiring two to four hops. Its construction preserves dependencies among intermediate questions and supporting passages, while filtering examples that can be answered through shortcuts [4]. This structured relation between intermediate questions and evidence is also useful when studying how generated decompositions divide responsibility for retrieving the evidence.

The three benchmarks therefore provide distinct operational views of compositional difficulty. They form the empirical basis of the layer-wise probing study introduced in Chapter 3 [5], without assuming that a nominal hop label is a complete measure of all forms of difficulty.

2.1.4 Evidence and Supporting Passages

Let the gold evidence for a question be

$$E(q) = \{e_1, e_2, \ldots, e_m\}, \tag{2.5}$$

and let the retriever produce the ordered ranking

$$R(q) = (d_1, d_2, \ldots, d_N). \tag{2.6}$$

For $e \in E(q)$, $\mathrm{rank}(e \mid q)$ denotes its one-based position in this ranking. Evidence retrieval is successful only to the extent that the required supporting passages appear within the part of the ranking made available downstream.

Answer correctness and evidence retrieval success are related but not equivalent. An LLM can occasionally answer correctly from parametric memory even when the annotated supporting passages are absent. Conversely, retrieving the gold passages does not guarantee that the generator will combine them correctly. For this reason, the later evaluation isolates whether required evidence is available instead of inferring retrieval quality solely from the final answer.

2.1.5 Evidence Coverage and Retrieval Depth

For evidence-oriented evaluation, a question is considered completely covered only when all of its annotated supporting evidence is available. The quantity of interest is therefore not how much of $E(q)$ a ranking contains, but how deep the ranking must be read before it contains all of it. For a question whose evidence is all present, that depth is fixed by the lowest-ranked required passage:

$$K^\star(q) = \max_{e \in E(q)} \operatorname{rank}(e \mid q). \tag{2.7}$$

Equivalently, $K^\star(q)$ is the smallest cutoff k at which $E(q) \subseteq D_k(q)$. If any required passage falls outside the evaluated search horizon, $K^\star(q)$ is undefined and the question is *censored*: assigning it an artificial finite depth would make a retrieval failure appear efficient.

Depth is a per-question quantity, so a set of questions is summarised by the proportion whose evidence is fully available by a given depth:

$$\text{Coverage}(k) = \frac{\left|\{\, q \in Q : \; K^\star(q) \le k \,\}\right|}{|Q|}. \tag{2.8}$$

A value of 0.66 at k means that two thirds of the questions have all of their evidence within the first k results, not that two thirds of the evidence has been found. Censored questions are counted in the denominator and never removed from it. Dropping them would evaluate each retrieval strategy on whichever subset of questions it happened to succeed on, and shallower depths would be obtained simply by failing more often.

Coverage(k) is non-decreasing in k, so it can be read in the other direction as well: the shallowest depth at which a given proportion α of the questions is served is

$$k_\alpha = \min\{\, k : \text{Coverage}(k) \ge \alpha \,\}. \tag{2.9}$$

Figure 2.3 illustrates both levels. On the left, one question's ranked list has its three gold passages at ranks 2, 5 and 7, so $K^\star = 7$: reading only the top five leaves this question uncovered even though two of its three passages have appeared. On the right, the depths of a cohort of questions accumulate into the coverage curve, from which k_α is read off.

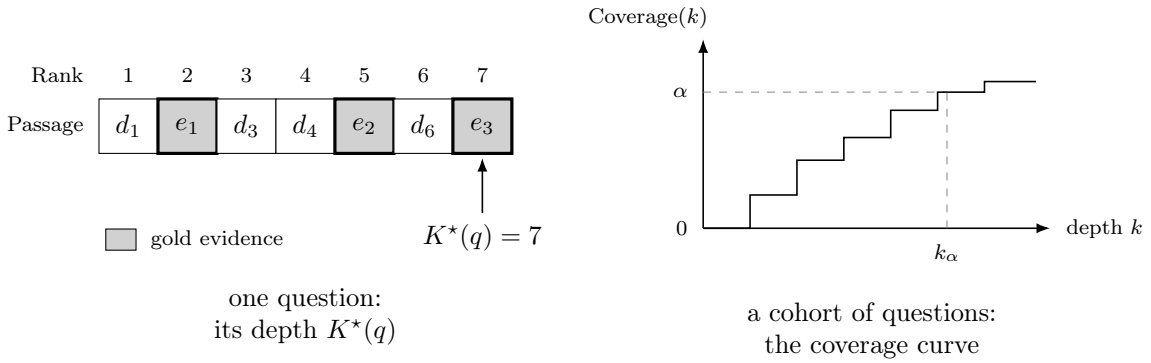


Figure 2.3: The two levels of the measurement. Left: for one question, $K^\star(q)$ is the rank of its lowest-ranked gold passage; reading the top five would leave it uncovered. Right: over a set of questions those depths accumulate into Coverage(k), the proportion covered by depth k, from which the depth k_α serving a proportion α is read. The curve is schematic.

Depth here is a rank-based proxy for retrieval effort. It is neither a reasoning hop count nor a measurement of wall-clock computation, which also depends on indexing, model architecture, batching, and hardware.

Coverage and depth must therefore be read together. A useful retrieval strategy should leave few questions uncovered and should place the evidence of the rest near the top of its rankings: at equal coverage, lower depth is preferable; at equal depth, higher coverage is preferable. *Coverage* in this sense means the recovery of annotated supporting passages, and is unrelated to *conformal* or *statistical* coverage, which concerns the long-run validity of prediction sets in uncertainty quantification.

2.2 Retrieval for Multi-Hop Reasoning

Retrieval methods can be characterised along two complementary dimensions. The first concerns how relevance between a query and a candidate passage is estimated through a score $s(q, d)$: sparse, dense, and late-interaction methods provide different scoring mechanisms, while rerankers refine an initial candidate ordering. The second concerns how evidence discovery is organised. A conventional retriever performs a single search using the original query, whereas iterative and graph-based approaches use intermediate state or structured relations to guide subsequent discovery. These dimensions are not mutually exclusive. An iterative retriever may use dense scoring internally, a graph-based architecture may use lexical or dense similarity to locate and score nodes or passages, and a first-stage sparse or dense search may be followed by reranking. Of the mechanisms described below, lexical and late-interaction methods are used directly in the later experiments for different purposes. BM25 provides the retrieval rankings used in the cross-benchmark budget-allocation studies of Chapters 4 and 5, while ColBERT similarities contribute features to the evidence-ownership estimator of Chapter 4. The remaining methods provide the broader retrieval context for the systems studied in this report. The remainder are presented because they define the vocabulary in which retrieval behaviour is described throughout this report.

2.2.1 Sparse Retrieval

Sparse retrieval represents text using high-dimensional term vectors. In a Term Frequency–Inverse Document Frequency (TF–IDF) representation, the weight of term t in document d is

$$w(t, d) = \mathrm{tf}(t, d)\,\mathrm{idf}(t), \qquad \mathrm{idf}(t) = \log \frac{N}{\mathrm{df}(t)}, \tag{2.10}$$

where $\mathrm{tf}(t, d)$ is the term frequency and $\mathrm{df}(t)$ counts documents containing t. Query and document vectors can then be compared with a cosine-style similarity. TF–IDF assigns greater influence to terms that are frequent in a document but rare throughout the corpus [6].

BM25 refines lexical scoring by saturating term-frequency gains and normalising for document length [7]:

$$\mathrm{BM25}(q, d) = \sum_{t \in q} \mathrm{IDF}(t) \frac{f(t, d)(k_1 + 1)}{f(t, d) + k_1 \left(1 - b + b \frac{|d|}{\mathrm{avgdl}}\right)}. \tag{2.11}$$

Here $f(t, d)$ is the term frequency, k_1 controls saturation, b controls document-length normalisation, and avgdl is average document length. Sparse indices are efficient and interpretable, and exact entity names often provide strong signals. Their principal limitation is lexical mismatch: a relevant passage can express the needed relation without sharing the query's wording, especially when the original question hides an intermediate entity.

2.2.2 Dense Retrieval

Dense retrieval learns encoders that map text into a continuous vector space. Dense Passage Retrieval (DPR), for example, uses separate question and passage encoders [8]:

$$\mathbf{q} = f_q(q), \qquad \mathbf{d} = f_d(d), \qquad s(q, d) = \mathbf{q}^\top \mathbf{d}. \tag{2.12}$$

Passage vectors can be precomputed and searched with approximate nearest-neighbour methods. Because the representation is learned, semantically related text can match despite limited word overlap. This is valuable when a question and its evidence use different expressions.

The compact representation also creates a bottleneck. A single vector must compress all potentially relevant entities and relations in a passage. For a complex query that simultaneously implies several information needs, the dot product provides only a coarse aggregate match and may obscure which part of the query aligns with which part of a passage.

2.2.3 Late-Interaction Retrieval and Reranking

Contextualized Late Interaction over BERT (ColBERT) occupies a middle ground between lexical matching and single-vector dense retrieval [9]. It independently encodes contextual query tokens $\mathbf{q}_i$ and document tokens $\mathbf{d}_j$, then applies a maximum-similarity operation for every query token:

$$s_{\text{ColBERT}}(q, d) = \sum_i \max_j \mathbf{q}_i^\top \mathbf{d}_j. \tag{2.13}$$

This *MaxSim* interaction preserves fine-grained matches while allowing document representations to be precomputed, which makes it applicable wherever a decision must be made about a specific query–passage pair rather than about a whole ranking.

Retrieval systems also commonly separate inexpensive candidate generation from more expensive scoring. A fast retriever first produces a manageable candidate set; a cross-encoder or another stronger reranker jointly examines each query–passage pair and reorders that set. Reranking can improve precision near the top but cannot recover a gold passage omitted from the candidate set. The initial retrieval stage therefore continues to constrain attainable evidence coverage.

2.2.4 Iterative and Multi-Hop Retrieval

Beyond the relevance-scoring mechanism itself, multi-hop questions raise the question of how retrieval should evolve as new information becomes available. Single-pass retrieval uses the same original question to search for every supporting passage, whereas iterative retrieval allows evidence acquired at one step to change the next query. Multi-Hop Dense

Retrieval (MDR) recursively retrieves passages with a dense representation conditioned on the question and earlier evidence, enabling later searches to incorporate newly identified bridge information [10].

Baleen addresses the rapid growth of the multi-hop search space through *condensed retrieval*: after each iteration, salient information from the retrieved material is condensed into a compact context for subsequent search. It combines this mechanism with focused late interaction and a learned ordering of retrieval steps [11].

Interleaving Retrieval with Chain-of-Thought Reasoning (IRCoT) alternates the two operations. An intermediate reasoning step specifies what information is needed next; retrieved evidence then updates the context used to produce the following step [12]. These approaches differ architecturally, but share a crucial observation: retrieval can benefit from intermediate state rather than repeatedly treating the initial question as a fixed search query.

2.2.5 Graph-Based Retrieval for Multi-Hop Reasoning

Iterative approaches organise evidence discovery through successive search steps, whereas graph-based approaches explicitly model relations among entities, passages, or concepts and use those relations to expose evidence that may be weakly connected to the original query. This is another way of organising retrieval, not a mutually exclusive alternative to sparse, dense, or late-interaction scoring: those mechanisms can still locate starting nodes or score candidate nodes and passages within a graph-based architecture. In multi-hop questions, a later passage may become relevant mainly through an entity or concept introduced by earlier evidence. Graph traversal or relevance propagation can make this connection explicit and expand beyond the passages found directly from the initial query.

HippoRAG is a representative approach. It builds a corpus-level knowledge graph, uses query–node similarity to identify starting points, and applies Personalized PageRank to propagate relevance toward connected knowledge and its source passages [13]. HippoRAG 2 retains this graph-propagation foundation while deepening the integration of passages and using an LLM more effectively during online retrieval [14]. The two systems illustrate how graph structure can help surface evidence connected through bridge entities even when that evidence is not retrieved directly from the original wording.

Graph-based retrieval therefore complements the sparse, dense, late-interaction, and iterative methods discussed above: it supplies an explicit mechanism for following associations across evidence items. What it does not do, and what none of the mechanisms in this section does, is determine which intermediate information need should be queried.

Because the depth required for complete evidence recovery varies substantially from one question to another, applying a uniformly expensive retrieval policy is neither necessary nor efficient. This observation leads to systems that estimate what an input needs and adapt their computation accordingly.

2.3 Difficulty-Aware and Adaptive Computation

Fixed pipelines allocate the same retrieval and reasoning procedure to every input, although a direct factual question and a four-step compositional question have different needs. Adaptive systems instead obtain a signal about the input or current state and translate it into an action.

Adaptive-RAG demonstrates complexity-aware routing for question answering. A classifier estimates whether a question is best handled without retrieval, with single-step retrieval, or with an iterative strategy, thereby avoiding the cost of a uniformly complex pipeline [15]. In generic form,

$$q \longrightarrow \widehat{d}(q) \longrightarrow \pi_{\widehat{d}}, \tag{2.14}$$

where $\widehat{d}(q)$ is an estimated difficulty or complexity level and $\pi_{\widehat{d}}$ is the selected computational policy. The important point is not a particular number of routing classes, but that a pre-answer estimate can govern retrieval, reasoning, or decomposition before their full costs are paid.

Adaptation can also occur after generation has begun. Forward-Looking Active REtrieval augmented generation (FLARE) anticipates the next sentence, examines confidence in the predicted content, and triggers retrieval when low-confidence tokens indicate that more information is needed. The retrieved documents are then used to regenerate the sentence [16]. Self-RAG trains a model to produce reflection signals that control retrieval on demand and support critique of both evidence and generated text [17].

Both systems make the same structural move: a quantity computed during processing is read as a signal, and that signal licenses a change of behaviour—in these cases re-retrieval, or the decision to retrieve at all.

These systems adapt on different signals. Adaptive-RAG estimates how much work a question will demand before any of that work is done; FLARE and Self-RAG estimate how reliable the content produced so far is. Both drive the same repertoire of actions—retrieve, re-retrieve, escalate—which makes the two easy to conflate, and they are not interchangeable. *Difficulty* is the amount of reasoning a question structurally requires; *uncertainty* is the reliability of what the system has produced. The two vary independently: a difficult question may be answered confidently when all the necessary evidence is available, and an easy one may remain uncertain because retrieval failed.

Existing routing approaches establish that question complexity is actionable, and they estimate it from surface characteristics with a separate classifier. That raises a representational question: could the target model itself already encode this information internally?

Before such a signal can guide an action, one must establish whether it is present in the model's representation and can be read out reliably. Probing provides a methodology for asking that question independently of the action that may eventually consume the signal.

2.4 Probing Internal Representations of Language Models

Neural models transform an input through a sequence of hidden representations. Probing studies these representations after training, asking what information about the input or computation is accessible at a particular stage. It is an analysis methodology rather than an adaptive policy: a decoded signal may later support routing, retrieval, or another action, but the probe itself only tests whether the signal can be recovered from a frozen model.

2.4.1 Linear Probing and Representation Analysis

Let the hidden representation of question q at layer ℓ be

$$\mathbf{h}^{(\ell)}(q) \in \mathbb{R}^d. \tag{2.15}$$

Given a target property y, a probe learns a readout g and predicts

$$\widehat{y} = g\big(\mathbf{h}^{(\ell)}(q)\big)\,. \tag{2.16}$$

The language model is kept fixed while the readout is trained. Comparing readouts across layers can then reveal where a property becomes more or less accessible, without changing the representations being examined.

A linear probe intentionally restricts the readout to a simple affine map,

$$g\big(\mathbf{h}^{(\ell)}(q)\big) = \mathbf{W}\mathbf{h}^{(\ell)}(q) + \mathbf{b}, \tag{2.17}$$

followed, when needed, by a standard decision rule. Its primary purpose is not to construct the strongest possible classifier. Rather, it tests whether the property is *readily decodable* through a deliberately limited mapping. Alain and Bengio introduced this perspective by fitting linear classifiers to intermediate neural-network features and tracking how class information became linearly separable with depth [18]. In the context of LLMs, the linear representation hypothesis similarly proposes that some semantic or high-level properties correspond to directions or comparatively simple geometric structure in activation space [19]. This hypothesis motivates linear readouts without requiring a broader mechanistic account of every computation performed by the model.

Probe performance nevertheless has a limited interpretation. High accuracy shows decodability under the chosen data distribution and probe family; it does not by itself show that the model uses the decoded feature causally, or that the feature will generalise beyond the evaluation setting. Hewitt and Liang further show that an expressive probe can learn a task or exploit memorisable correlations even when the representation does not encode the intended structure in a meaningful way. They propose control tasks and selectivity as ways to put task accuracy in the context of probe capacity [20]. Probing evidence should therefore be accompanied by appropriate controls and interpreted as evidence of accessible representational structure, not as an unrestricted claim about the model's reasoning process.

2.4.2 Latent Properties in LLM Representations

Prior work indicates that language models contain signals associated with knowledge and confidence. Kadavath et al. show that models can assess whether their proposed answers are likely to be correct and can estimate whether they know an answer, establishing a behavioural form of model self-knowledge [21]. Although this self-evaluation is not itself the same as hidden-state probing, it motivates the search for related information inside the representations that precede an answer.

Several studies recover truth-related structure directly from activations. Burns et al. identify latent knowledge without labelled examples by exploiting consistency between statements and their negations [22]. Marks and Tegmark find comparatively simple geometric structure separating true and false statements and study the transfer of truth directions across datasets [23]. Azaria and Mitchell likewise train classifiers on internal activations to distinguish truthful from false statements [24]. Together, these results

show that semantic properties such as truth can be accessible to relatively simple read-outs, while also reinforcing the need to test whether an apparent direction transfers and captures the intended property.

Question-only probing moves closer to the setting of this thesis. Moreno Cencerrado et al. extract a representation after a question has been read but before an answer is generated, and show that a linear probe can predict aspects of the eventual answer's correctness and confidence [25]. This establishes that a question representation may contain predictive information about later success. The target considered here is different: previous work largely asks what the model knows, whether a statement is true, or whether the eventual answer is likely to be correct, whereas this thesis asks about the *compositional difficulty of the input question*—how much multi-step reasoning the question demands.

Yang et al. examine a related but distinct issue: whether LLMs internally recall and compose intermediate facts while processing a multi-hop prompt [26]. Their question concerns the execution of a latent reasoning pathway. It should not be conflated with estimating the amount of reasoning demanded by the input:

$$\underbrace{\text{Is multi-hop reasoning executed internally?}}_{\text{latent reasoning}} \neq \underbrace{\text{How much reasoning is required?}}_{\text{compositional difficulty}}. \quad (2.18)$$

The former asks what occurs during a model's forward computation; the latter is the logically earlier question of whether the required degree of composition is already encoded in the question representation before answering.

Prior work therefore shows that internal LLM representations contain decodable information about knowledge, truthfulness, confidence, and eventual answer correctness. Considerably less attention has been given to whether the question representation itself encodes compositional difficulty before answer generation. The next section turns from this analysis methodology to question decomposition, one possible action when additional compositional reasoning is required.

2.5 Question Decomposition for Multi-Hop Reasoning

2.5.1 Why Decomposition Helps Multi-Hop Reasoning

A multi-hop question often conceals several intermediate information needs within a single surface form. Answering it may require identifying an entity, recovering a property of that entity, and only then applying a comparison or other operation. When these steps remain implicit, a model must determine the reasoning path, obtain the necessary facts, and compose them at the same time. Question decomposition instead converts the original question into simpler, explicit subproblems whose answers can be combined to obtain the final answer.

This separation is useful because knowing the answers to the component questions does not guarantee that a model can answer their composition. Press et al. describe this discrepancy as the *compositionality gap* and show that it is not necessarily removed by increasing model scale [27]. Their Self-Ask method narrows the gap by having the model generate and answer explicit follow-up questions before producing its final response. More generally, decomposition reduces a complex inference into smaller reasoning steps, makes dependencies between intermediate answers explicit, and allows an error to be associated with a particular subproblem rather than with the question as a whole.

Focused subquestions may also provide more effective retrieval queries than the original multi-hop question. Each subquestion expresses a narrower information need and can therefore be matched against evidence relevant to one step of the reasoning chain. Retrieval is not the primary objective of every decomposition method, but this property provides an important connection between explicit reasoning structure and multi-hop evidence retrieval.

2.5.2 Earlier Question-Decomposition Approaches

Question decomposition was studied before the emergence of modern instruction-tuned LLMs. These earlier approaches treated the decomposition as a learned intermediate representation between a complex input question and the single-hop models used to answer it. The intermediate subquestions made it possible to reuse existing question-answering components, but producing them typically required a task-specific architecture, training procedure, or form of decomposition supervision.

Min et al. decompose multi-hop reading-comprehension questions into simpler single-hop questions, answer the resulting questions with existing readers, and globally rescore candidate decompositions and answers. To limit the need for manually annotated decompositions, their approach casts subquestion generation as span prediction over the original question [28]. The resulting decomposition is therefore not merely an explanation appended to an answer; it is an intermediate structure on which the downstream answering process operates.

Perez et al. further reduce reliance on decomposition labels through an unsupervised one-to-many sequence-transduction model. Their system learns to map a multi-hop question to multiple single-hop subquestions, sends those subquestions to an off-the-shelf question-answering model, and recomposes the returned answers into a final prediction [29]. Together, these approaches established decomposition as a useful interface between complex questions and reusable single-hop components. At the same time, the decomposer remained a dedicated learned model designed for a particular decomposition and answering pipeline.

2.5.3 LLM-Based Decomposition

Large language models shifted question decomposition from a task-specific prediction problem toward a prompting problem. Rather than training a dedicated decomposer for each setting, a general-purpose model can be instructed—and, when useful, shown examples—to identify subproblems, solve them in an appropriate order, and combine their results. Self-Ask is an early example of this pattern: the prompt elicits follow-up questions whenever an intermediate fact is needed [27].

Successive Prompting interleaves decomposition and answering. At each step, the model generates a simpler question, answers it, and uses the accumulated context to determine the next question until the original problem is resolved [30]. Least-to-Most Prompting separates these phases more clearly: it first decomposes a complex problem into a sequence of simpler subproblems and then solves them from least to most complex, using earlier solutions to support later ones [31]. Decomposed Prompting adopts a modular formulation in which a controller prompt breaks a task into subtasks and delegates them to specialised prompts, models, or symbolic modules [32].

Although these methods differ in when decomposition occurs and how subproblems are

solved, they replace a fixed learned decomposer with behaviour elicited largely through prompting. Decomposition consequently becomes a configurable reasoning component: instructions, demonstrations, output formats, and available modules can change both the number and the form of the generated subquestions. This flexibility makes decomposition easier to adapt across tasks, while also making the quality of the prompt and of the generated intermediate structure consequential for all subsequent reasoning and retrieval.

2.6 Research Gap and Thesis Positioning

The preceding literature connects complex questions to adaptive computation, probing, and decomposition, but leaves two questions that structure the experimental chapters of this thesis.

2.6.1 Anticipating Compositional Difficulty

Adaptive computation benefits from an estimate of query difficulty before an expensive policy is selected. At the same time, prior probing work shows that knowledge, truth, confidence, and eventual answer correctness can be decoded from internal representations. Whether those representations also encode the compositional difficulty of the question itself remains unresolved and is the first research question of this thesis.

Does the LLM already encode how much compositional reasoning a question requires?

Chapter 3 investigates this question while keeping compositional difficulty conceptually distinct from expected answer correctness.

2.6.2 Optimising Decomposition for Retrieval Efficiency

Question decomposition exposes intermediate information needs, and iterative retrieval demonstrates that such intermediate state can improve search. The remaining issue is whether an arbitrary generated decomposition can be judged and improved specifically as a set of retrieval queries.

Can a decomposition be evaluated and improved so that its subquestions retrieve the required evidence more efficiently?

Answering this requires determining which evidence each subquestion is responsible for, evaluating decompositions jointly through evidence coverage and retrieval depth, and controlling how the available retrieval budget is used. Chapter 4 develops the evidence-aware evaluation and attribution required for these measurements. Chapter 5 then applies them to automated decomposition-instruction search and learned passage allocation, while also developing structured evidence feedback for subsequent prompt optimisation.

Conclusion

This chapter presented the background necessary for the experimental work in the remainder of the thesis. Multi-hop question answering requires both reasoning across several pieces of evidence and retrieval mechanisms capable of exposing that evidence efficiently. Existing work approaches these demands through improved retrieval, adaptive computation, and question decomposition, while probing provides a means of identifying internal signals that may support such adaptation.

Two questions remain central, and they approach the same difficulty from opposite sides: one asks whether the amount of reasoning a question requires is already visible inside the model, the other asks how the retrieval work that this amount implies can be reduced outside it. The first is whether compositional difficulty is encoded in the question representation before answering, which is investigated in Chapter 3. The second is whether decomposition can be evaluated and improved to preserve evidence coverage while using retrieval depth and passage budgets more effectively, which motivates Chapters 4 and 5.

Chapter 3
Probing Compositional Question Difficulty in Language Model Representations

Introduction

The preceding chapter established the importance of anticipating how much compositional reasoning a question may require before committing to a fixed retrieval or reasoning strategy. This chapter examines whether such difficulty is already reflected in the internal representations of a language model while it processes the question, before any answer is generated.[1]

Hidden states are extracted across the layers of three open instruction-tuned models and read with simple linear probes on three multi-hop benchmarks whose notions of difficulty differ in kind; every positive result is then examined through a series of controls.

3.1 Research Question and Problem Formulation

Let

$$\mathcal{D} = \{(q_i, y_i)\}_{i=1}^{N} \tag{3.1}$$

be a collection of questions, where q_i is a question and y_i is its difficulty label. Depending on the benchmark, y_i may be a binary easy–hard split, a human-assigned difficulty level, or a structural hop count. These labels provide operational proxies for compositional difficulty: they make the question measurable, but they do not constitute a universal definition of difficulty.

A question can require several compositional steps and still be answered correctly. The target here is its benchmark-defined difficulty, rather than the probability that a particular model fails or the retrieval depth needed to find its evidence.

For a frozen language model M, processing question q_i produces a representation

$$\mathbf{r}_i^{(\ell)} \in \mathbb{R}^d \tag{3.2}$$

[1] The study reported in this chapter was carried out during the internship and has been accepted for publication [5]. The present author is its first author and was responsible for the design and implementation of the probing pipeline, its validation controls, and the analysis of the results. The chapter develops the study in extended form.

at layer ℓ, where d is the hidden dimension. The principal research question is whether a simple function of $\mathbf{r}_i^{(\ell)}$ can recover y_i, and how that recoverability changes across model depth. More precisely, the experiments ask whether there exists a linear score

$$s_i^{(\ell)} = \mathbf{w}^{(\ell)\top}\mathbf{r}_i^{(\ell)} \tag{3.3}$$

that orders questions according to their benchmark-defined difficulty on examples not used to fit the direction $\mathbf{w}^{(\ell)}$. How that direction is obtained is the subject of Section 3.2.2.

Decodability is the weakest of the four properties established here. A representation that genuinely encodes compositional structure should also exceed what the same procedure yields when the labels are shuffled, survive the removal of the question's surface statistics, order the difficulty levels rather than merely separate them, and hold across benchmarks that define difficulty differently. The four results follow in that order:

1. difficulty is decodable from the question representation across model depth;
2. the observed performance exceeds a shuffled-label null and is not exhausted by six selected surface features;
3. the decoded direction orders an unseen intermediate difficulty class between easy and hard extremes; and
4. the direction, after controlling for surface features, transfers across benchmarks within the representation space of the same model.

Figure 3.1 summarises this logic. The language model is never trained or fine-tuned and no answer is generated: only the representations formed during the question's forward pass are studied.

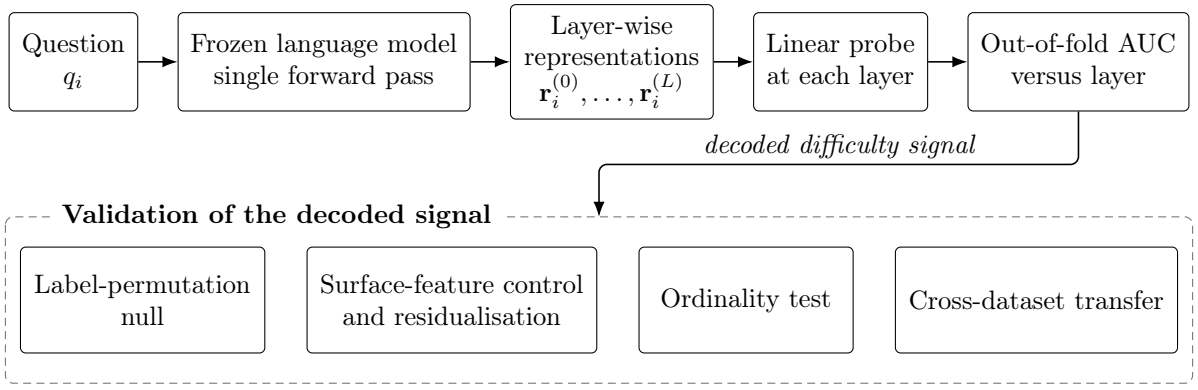


Figure 3.1: Overview of the compositional-difficulty probing protocol. A frozen language model processes each question once, and a linear probe is evaluated independently at each layer using out-of-fold predictions. The decoded signal is subsequently examined through permutation, surface-feature, ordinality, and cross-dataset transfer analyses.

3.2 Methodology

3.2.1 Hidden-State Extraction and Question Representation

Each question is tokenised as raw text, without a chat template, and passed through a frozen causal language model once. Hidden states are read from all $L + 1$ positions in

depth: layer 0 is the token-embedding output, while layers $1, \ldots, L$ are the outputs of the transformer blocks. Extraction is forward-pass only; the model does not generate an answer.

Because questions contain different numbers of tokens, each layer's token-level states must be reduced to one fixed-dimensional question vector. Let $\mathbf{h}_{it}^{(\ell)}$ be the hidden state of token t for question i at layer ℓ, and let $m_{it} \in \{0, 1\}$ denote its attention mask. The reported experiments use mask-aware last-token pooling:

$$t_i^\star = \max\{t : m_{it} = 1\}, \qquad \mathbf{r}_i^{(\ell)} = \mathbf{h}_{it_i^\star}^{(\ell)}. \tag{3.4}$$

The maximum in Equation (3.4) identifies the final non-padding token, ensuring that padding is never selected. Last-token pooling is used because the model is causal: the final non-padding position is the only one whose representation has attended to the whole question. Mean pooling over the question's tokens is an equally defensible choice and is not evaluated here. For each question, the complete extracted representation is therefore

$$\mathbf{R}_i = \begin{bmatrix} \mathbf{r}_i^{(0)} \\ \mathbf{r}_i^{(1)} \\ \vdots \\ \mathbf{r}_i^{(L)} \end{bmatrix} \in \mathbb{R}^{(L+1)\times d}. \tag{3.5}$$

Keeping one vector per layer makes it possible to determine not only whether difficulty is decodable, but also where in the network the signal emerges and where it is strongest.

3.2.2 Linear Probe and Layer-Wise Evaluation

The readout is deliberately the simplest one that can still order questions. At a given layer it averages the representations of the difficult questions, averages those of the easy questions, and takes the difference between the two averages; a new question is then scored by how far along that direction it falls. Formally, this is a regularisation-free diagonal Linear Discriminant Analysis (LDA) probe. Its simplicity is the point: the activation dimension d is several thousand while the number of questions N is at most 900, and a classifier with more freedom could fit the labels in that regime without reading anything from the representation.

All preprocessing is fitted within the training part of each fold. For feature j, the standardised activation is

$$\widehat{r}_{ij}^{(\ell)} = \frac{r_{ij}^{(\ell)} - \mu_j^{(\ell)}}{\sigma_j^{(\ell)}}, \tag{3.6}$$

where the mean $\mu_j^{(\ell)}$ and standard deviation $\sigma_j^{(\ell)}$ are estimated only from the training fold. For a binary contrast, let $\overline{\mathbf{r}}_+$ and $\overline{\mathbf{r}}_-$ be the two class means in standardised space. The probe direction and score are

$$\mathbf{w} = \overline{\mathbf{r}}_+ - \overline{\mathbf{r}}_-, \qquad s_i = \mathbf{w}^\top \widehat{\mathbf{r}}_i. \tag{3.7}$$

The direction is therefore fixed entirely by the two class means: the probe has no weights to tune and no other free parameters.

The same construction covers the benchmarks that have three difficulty levels. It is applied once per level: that level takes the role of the difficult class and the other two together take the role of the easy one, which gives one direction for each level.

Evaluation uses five-fold cross-validation. Each question receives a score from a probe that was not trained on that question, and the pooled out-of-fold scores are used to compute the Area Under the Receiver Operating Characteristic Curve (AUC). AUC has a direct reading in this setting: it is the probability that a randomly chosen difficult question receives a higher score than a randomly chosen easy one, so 0.5 corresponds to chance and 1.0 to a perfect ordering. The procedure is repeated independently at every layer to form an AUC-versus-layer curve. Where there are three levels, each of those directions is scored separately and the three AUCs are averaged; this average is what the macro one-versus-rest AUC means in the tables that follow. The binary 2Wiki contrast has a single direction and a single AUC. AUC is the primary measure because it evaluates ranking without requiring a deployment threshold.

3.2.3 Label-Permutation Null

A probe is fitted at every one of several dozen layers, using several thousand features each time. A high value somewhere in that grid could therefore appear even when questions and labels have no relation at all. The label-permutation control measures how large such a value can be when that relation is removed by construction. The complete probing procedure is repeated for $P = 200$ random shuffles of the labels, yielding a null distribution at each layer.

The empirical AUC curve is compared with the null's 5th–95th percentile band. If the real curve lies well above that band, its performance cannot be attributed to the capacity of this probing procedure under randomly associated labels. This is a statistical control for decodability; it does not, by itself, rule out meaningful but superficial correlates of the true labels.

3.2.4 Surface-Feature Control and Residualisation

Benchmark difficulty may be correlated with visible properties of the question. Longer questions, for instance, may tend to contain more relations. The study collects six inexpensive surface statistics in $\boldsymbol{\phi}_i$: character length, word count, capitalised-word count as a named-entity proxy, digit-token count, punctuation count, and tokenised length. A surface-only probe establishes how predictive these statistics are on their own.

The residual test asks what is left in the hidden state once those cues have been taken out of it. For every activation feature, the part of its value that can be predicted from $\boldsymbol{\phi}_i$ is estimated and subtracted, leaving only the part the six statistics cannot account for. The prediction is fitted inside each training fold, giving coefficients $\mathbf{B}^{(\ell)}$; both training and held-out examples are then transformed as

$$\widetilde{\mathbf{r}}_i^{(\ell)} = \mathbf{r}_i^{(\ell)} - \mathbf{B}^{(\ell)\top}\boldsymbol{\phi}_i. \tag{3.8}$$

The linear probe is then fitted to these residualised representations. Because the regression coefficients and all subsequent preprocessing are estimated inside the training fold, the held-out labels and activations do not influence the transformation used to score them.

The comparison has three parts: the raw activation probe measures total decodability, the surface-only probe measures the selected visible cues, and the residual probe measures the component not linearly explained by those cues. A positive residual result supports a component beyond the six selected features; it does not prove complete independence from every possible aspect of surface form.

3.2.5 Ordinality Test

For benchmarks with at least three ordered classes, class separation need not imply a graded representation. A classifier might distinguish unrelated templates for each label without arranging them along one difficulty axis. The ordinality test therefore fits $\mathbf{w}$ using only the lowest and highest difficulty classes, then projects the unseen middle class onto that direction.

Let $\overline{s}_{\text{low}}$, $\overline{s}_{\text{mid}}$, and $\overline{s}_{\text{high}}$ be the out-of-fold mean projection scores for the three classes. Their interpolation fraction is

$$\rho^{(\ell)} = \frac{\overline{s}^{(\ell)}_{\text{mid}} - \overline{s}^{(\ell)}_{\text{low}}}{\overline{s}^{(\ell)}_{\text{high}} - \overline{s}^{(\ell)}_{\text{low}}}. \tag{3.9}$$

A value near 0 places the middle class close to the easy extreme, a value near 1 places it close to the hard extreme, and $0 < \rho^{(\ell)} < 1$ places it between the two. The central point is that the middle class is never used to define the direction; interpolation therefore tests an ordering that the probe was not trained to impose.

For MuSiQue, the extremes are 2-hop and 4-hop questions and the held-out middle class is 3-hop. For HotpotQA, the easy and hard labels define the axis and the medium class tests interpolation. The binary 2Wiki split cannot support this three-level test.

3.2.6 Cross-Dataset Transfer

The final validation asks whether a decoded direction captures a property that is shared across benchmarks rather than their individual topics or templates. A direction fitted on the extreme classes of source benchmark A is applied to the extreme classes of target benchmark B, within the activation space of a fixed language model. Transfer is summarised by the retention ratio

$$R^{(\ell)}_{A\to B} = \frac{\text{AUC}^{(\ell)}_{A\to B} - \frac{1}{2}}{\text{AUC}^{(\ell)}_{B} - \frac{1}{2}}, \tag{3.10}$$

where $\text{AUC}_{A\to B}$ is obtained by training on A and testing on B, and AUC_B is the target's own within-benchmark, cross-validated extreme-class score. Chance is 0.5, so subtracting it from both terms compares the two scores by how far each rises above chance rather than by their raw values. A retention near 1 means the transferred direction recovers what a probe trained on the target benchmark itself achieves; $R = 0$ corresponds to chance transfer and $R < 0$ to below-chance transfer.

The reported principal analysis uses a fused leave-one-benchmark-out variant: for each target, the extreme classes of the other two benchmarks are pooled to fit the axis, which is then evaluated on the held-out benchmark's extreme classes. AUC and retention are averaged over the informative layer band in which the target benchmark's own extreme-class probe performs most strongly. Selecting this band from the target's within-benchmark curve, rather than from the transfer curve, prevents the reported transfer from being chosen at its own maximum.

Transfer here is *cross-benchmark*, not cross-model. Every transferred axis is trained and tested on representations produced by the same language model. Whether an axis learned in one model can be applied directly to another model is outside the scope of the experiment.

3.3 Experimental Setup

3.3.1 Language Models

The experiments cover three open instruction-tuned language models: Qwen2.5-32B, Mistral-24B, and Gemma-27B. They have 65, 41, and 47 hidden-state positions respectively, counting the embedding output as position 0. Each model is probed only on the activations that it produced itself. This design supports replication of the finding across three model families, but it does not evaluate transfer of a probe between models.

3.3.2 Benchmarks and Difficulty Labels

Three English, Wikipedia-based multi-hop question-answering benchmarks provide deliberately different difficulty definitions:

- **2WikiMultiHopQA** [3] supplies a binary contrast derived from evidence-chain length. Questions with at most two evidence items form the easy class, and questions with at least four form the hard class.
- **HotpotQA** [2] provides human-annotated easy, medium, and hard difficulty levels.
- **MuSiQue** [4] supplies a structural composition label of 2-hop, 3-hop, or 4-hop.

The analysis samples 300 questions per class, producing 600 questions for the binary 2Wiki task and 900 each for HotpotQA and MuSiQue. The class-balanced design prevents class prevalence from dominating the evaluation. More importantly, the variation in label type makes agreement across benchmarks stronger evidence than repeated evaluation on one construction scheme. It also requires a conservative interpretation: evidence-chain length, human judgement, and hop count are related proxies, not interchangeable definitions of an abstract latent quantity.

3.3.3 Evaluation Configuration

The procedure of Section 3.2 is applied unchanged to all nine model–benchmark pairs. Two reporting conventions hold throughout. Confidence intervals are the half-width of a 95% bootstrap interval over questions, from 2,000 resamples; the reference quoted from the label-permutation null is its 95th percentile over the 200 shuffles.

Unless stated otherwise, the figures use Qwen2.5-32B on MuSiQue as a running example, while the tables report all nine model–benchmark combinations. These choices keep the layer-wise plots readable without reducing the breadth of the numerical comparison.

3.4 Results

3.4.1 Layer-Wise Decodability of Compositional Difficulty

Table 3.1 shows that compositional difficulty is linearly decodable above chance for every model–benchmark pair. Structural labels are especially separable: the 2Wiki binary contrast reaches an AUC from 0.972 to 0.987, and MuSiQue's three-class hop count reaches

Table 3.1: Difficulty is linearly decodable from question representations beyond the selected surface features. N is the sample size, C the number of classes, and Layer the best-performing hidden-state position, with embeddings at 0. Probe AUC is macro one-versus-rest AUC, except for binary 2Wiki. "Surf. resid." is the peak after fold-local surface residualisation; Null95 is the 95th percentile of the 200-run permutation null. The $\pm$ value is the half-width of a 95% bootstrap interval over questions using 2,000 resamples.

Model	Benchmark	N	C	Layer	Probe AUC	Surf. only	Surf. resid. / Null95
Qwen2.5-32B	2Wiki	600	2	64	0.987 ± 0.006	0.907	0.838 ± 0.034 / 0.55
	HotpotQA	900	3	44	0.716 ± 0.024	0.661	0.627 ± 0.027 / 0.53
	MuSiQue	900	3	24	0.896 ± 0.013	0.779	0.774 ± 0.022 / 0.53
Mistral-24B	2Wiki	600	2	21	0.972 ± 0.010	0.911	0.829 ± 0.035 / 0.55
	HotpotQA	900	3	18	0.694 ± 0.025	0.660	0.604 ± 0.027 / 0.53
	MuSiQue	900	3	12	0.874 ± 0.015	0.776	0.753 ± 0.023 / 0.53
Gemma-27B	2Wiki	600	2	44	0.978 ± 0.010	0.909	0.833 ± 0.035 / 0.54
	HotpotQA	900	3	23	0.709 ± 0.024	0.661	0.606 ± 0.027 / 0.53
	MuSiQue	900	3	24	0.880 ± 0.015	0.780	0.766 ± 0.023 / 0.53

a macro AUC from 0.874 to 0.896. HotpotQA's human-labelled levels are less separable, with AUC from 0.694 to 0.716, consistent with a noisier and less purely structural proxy.

The layer profile is as important as the peak. Figure 3.2 shows that Qwen2.5-32B is at chance at the embedding layer on MuSiQue, after which decoding performance rises through the transformer and peaks at layer 24. The same emergence profile occurs in all nine settings: the signal is absent from the token embeddings and becomes decodable as the question is processed. Where it peaks, however, tracks the benchmark's notion of difficulty more closely than it tracks the model. The structural hop count of MuSiQue is decoded earliest, at roughly a third to a half of network depth: layer 24 of 65 for Qwen2.5-32B, layer 12 of 41 for Mistral-24B, and layer 24 of 47 for Gemma-27B. HotpotQA's human annotation peaks nearer the middle, and the 2Wiki evidence-chain contrast peaks latest, at or close to the final layer in two of the three models. What replicates across settings is therefore the emergence of the signal with depth, not one preferred layer.

This trajectory supports a representational interpretation stronger than a single high peak: the final token's contextual representation acquires difficulty-related structure across depth. It remains a statement about what is linearly readable, however, not proof that subsequent model computation uses that structure.

3.4.2 Robustness Beyond Surface Features

The first control verifies that the decoding curve is not reproduced when difficulty labels are detached from their questions. Across settings, the 95th-percentile permutation reference remains around 0.53–0.55, far below the observed peaks in Table 3.1. Figure 3.3 shows the complete 200-shuffle null band for the Qwen2.5-32B–MuSiQue example. The empirical curve rises well above the band over a broad span of layers rather than at one isolated position.

Surface form nevertheless explains a meaningful part of performance. A probe using only the six surface statistics reaches 0.907–0.911 on 2Wiki and 0.776–0.780 on MuSiQue, so the raw result cannot be described as independent of visible question properties. The

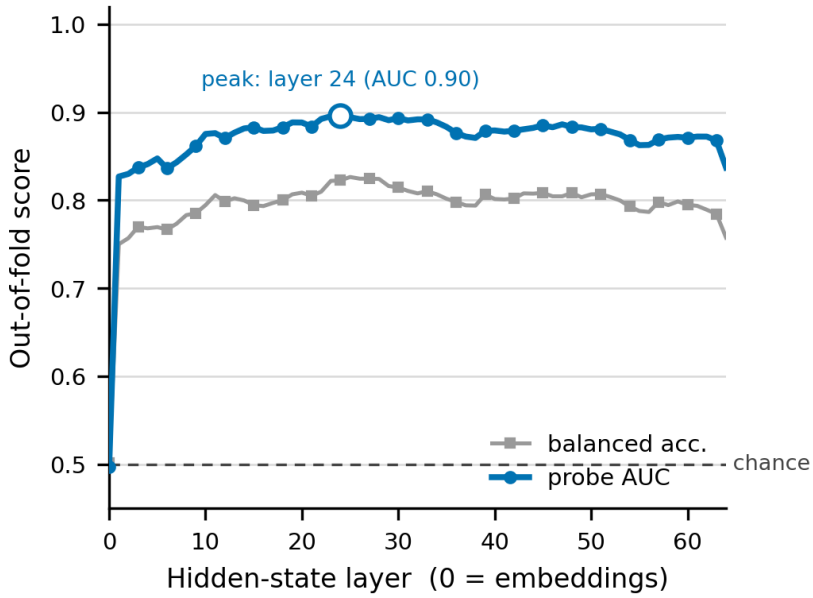


Figure 3.2: Difficulty becomes linearly decodable as the question is processed. The figure reports out-of-fold macro one-versus-rest AUC and balanced accuracy by hidden-state position for a diagonal-LDA probe on Qwen2.5-32B and MuSiQue. Performance begins near chance at the embedding representation and peaks in the intermediate layers. Adapted from [5].

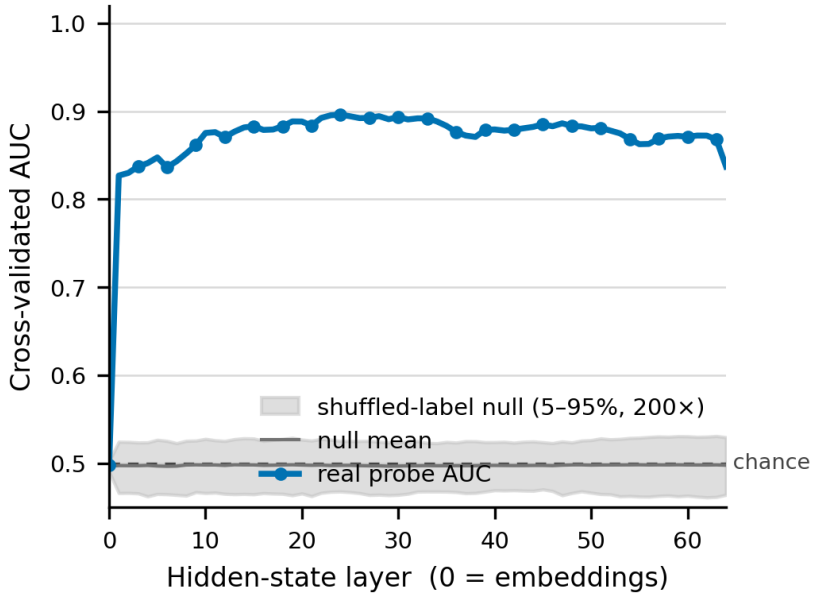


Figure 3.3: The real probe curve lies well above the 5th–95th percentile band obtained from 200 shuffled-label runs for Qwen2.5-32B on MuSiQue. The control shows that the observed layer-wise decodability is not reproduced by the probing pipeline when question–label correspondence is destroyed. Adapted from [5].

residual analysis asks the narrower and more defensible question: does a decodable component remain after removing the linear contribution of those selected properties?

The answer is positive for all model–benchmark pairs. Residual peak AUC is 0.829–0.838 on 2Wiki, 0.753–0.774 on MuSiQue, and 0.604–0.627 on HotpotQA, all above the corresponding null reference. Figure 3.4 shows that the residual curve for Qwen2.5-32B on MuSiQue remains elevated across depth even though it is lower than the raw curve.

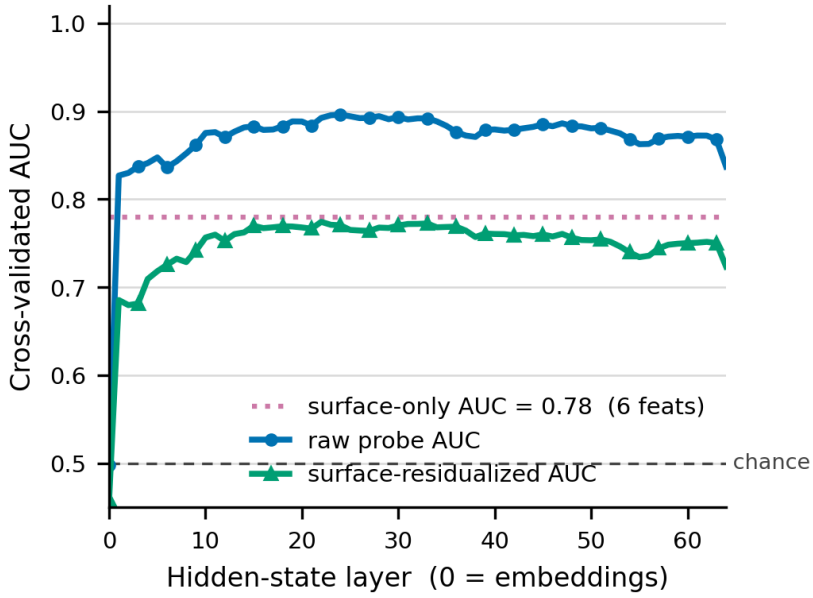


Figure 3.4: Raw and surface-residualised probe AUC across depth, with the surface-only score as a reference, for Qwen2.5-32B on MuSiQue. Removing six selected surface features reduces performance but leaves a substantial above-null signal. This supports a component beyond those features, not complete independence from all surface form. Adapted from [5].

Together, the two controls delimit the claim. The signal is not a shuffled-label artefact, and the six measured surface statistics do not exhaust it. At the same time, the strength of surface-only prediction shows that compositional difficulty is partly expressed in the wording and structure of the input.

3.4.3 Ordinal Structure of the Difficulty Direction

The extreme-class probes separate low from high difficulty almost perfectly on MuSiQue, reaching AUC values up to 0.99. More importantly, the unseen 3-hop class falls between the 2-hop and 4-hop extremes. Its interpolation fraction is 0.58 for Qwen2.5-32B, 0.56 for Mistral-24B, and 0.47 for Gemma-27B. These values cluster around the midpoint despite the fact that 3-hop questions never define the direction.

Figure 3.5 displays both the strength of the extreme-class contrast and the position of the held-out class. HotpotQA exhibits the same ordering, although its medium class lies closer to the hard extreme, with interpolation fractions from 0.93 to 0.96. The result therefore does not claim equal spacing among labels. It shows that the representation arranges the middle class in the expected order along a direction learned only from the endpoints, which is evidence for a graded difficulty axis rather than three disconnected class templates.

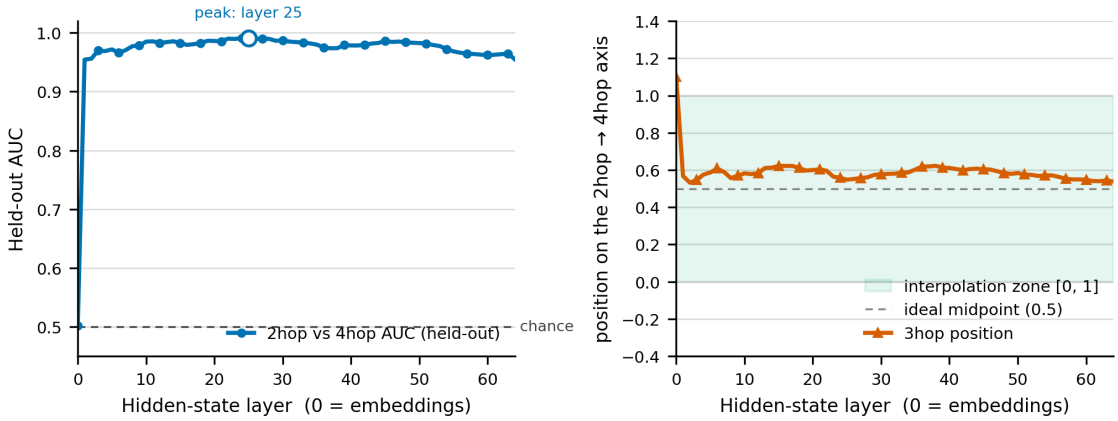


Figure 3.5: Left: out-of-fold AUC for the 2-hop versus 4-hop contrast by hidden-state position. Right: the interpolation fraction of the held-out 3-hop class. Across informative depths, the intermediate class lies between the two extremes it was not used to fit. Results are for Qwen2.5-32B on MuSiQue. The peak differs from the three-class peak because this test uses a binary extreme-class contrast. Adapted from [5].

3.4.4 Cross-Dataset Transfer of the Residualised Difficulty Direction

Table 3.2 presents the fused leave-one-benchmark-out analysis. The raw axis transfers unevenly. It retains 0.54–0.71 of above-chance within-benchmark performance on 2Wiki and only 0.25–0.39 on MuSiQue; on HotpotQA, raw transfer is below chance, producing negative retention from −0.48 to −0.42. This failure is consistent with the raw axis mixing compositional structure with dataset-specific surface cues. Because retention is a ratio to each benchmark's own within-benchmark reference, and that reference is weak for HotpotQA, values above 1 occur; their interpretation is given after the table.

After surface residualisation, transfer becomes strong in all nine cases. Retention rises to 0.80–0.98 for 2Wiki, 1.02–1.29 for MuSiQue, and 1.95–2.44 for HotpotQA. Figure 3.6 illustrates the Qwen2.5-32B transfer-to-MuSiQue case: the axis is fitted on the pooled extreme classes of 2Wiki and HotpotQA, then evaluated on MuSiQue.

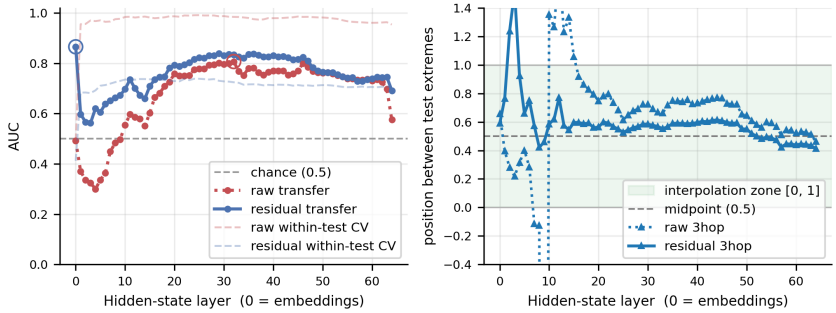


Figure 3.6: Transfer to MuSiQue in the representation space of Qwen2.5-32B, with the difficulty direction fitted on pooled extreme classes from 2Wiki and HotpotQA. The raw direction transfers weakly, whereas surface residualisation yields strong transfer across the informative layer band. This is cross-benchmark transfer within one model, not transfer between model families. Adapted from [5].

Table 3.2: Fused cross-benchmark transfer within each model. For each test benchmark, the axis is fitted on pooled extreme classes from the other two benchmarks. AUC is the mean binary extreme-class score over informative layers, and $R = (\text{AUC} - 0.5)/(\text{AUC}_{\text{within}} - 0.5)$ is retention relative to the test benchmark's own within-benchmark score. Raw uses activations directly; Resid. first removes the six surface features. These binary extreme-class AUCs are not directly comparable to the three-class macro peaks in Table 3.1.

		Raw		Residualised	
Model	Test benchmark	AUC	R	AUC	R
Qwen2.5-32B	2Wiki	0.823	0.71	0.796	0.94
	HotpotQA	0.384	-0.43	0.774	1.95
	MuSiQue	0.685	0.39	0.769	1.22
Mistral-24B	2Wiki	0.774	0.61	0.779	0.98
	HotpotQA	0.379	-0.48	0.764	2.21
	MuSiQue	0.627	0.27	0.720	1.02
Gemma-27B	2Wiki	0.744	0.54	0.724	0.80
	HotpotQA	0.398	-0.42	0.756	2.44
	MuSiQue	0.610	0.25	0.768	1.29

A retention greater than 1 does not mean that more than 100% of difficulty has been recovered. Retention normalises the transferred AUC above chance by a within-benchmark reference. HotpotQA has a comparatively weak residual within-benchmark signal, so a cleaner axis pooled from two other benchmarks can exceed that reference and produce a large ratio. The central finding is the consistent direction of the residual transfer, including between a human difficulty annotation and structural hop-based labels. Within each model's representation space, the component beyond the selected surface features is therefore substantially more benchmark-general than the raw axis.

3.5 Discussion and Limitations

3.5.1 Interpretation and Implications for Adaptive Computation

The experiments support a coherent representational account. Difficulty is not linearly available in the embedding-layer representation of the last input token, but becomes decodable as contextual processing proceeds; where it is strongest depends on the benchmark's notion of difficulty, from roughly a third of network depth for hop counts to the final layers for the evidence-chain contrast. The decoded direction also has ordinal structure: it places an unseen middle class between extremes. Finally, surface residualisation weakens the within-benchmark signal yet makes its cross-benchmark behaviour markedly more consistent. Taken together, these results suggest that the model forms a graded representation related to how much compositional reasoning a question requires.

The comparison with the surface-only probe deserves a direct statement, since those six statistics are themselves strong predictors. Read as a predictor, the probe is the better of the two in all nine settings, by between 0.03 and 0.12 AUC. Read as a signal, the gap

is wider than that margin suggests. The raw axis, which still carries surface information, transfers poorly between benchmarks and inverts on HotpotQA, whereas the residualised axis transfers in every case. Surface statistics are therefore a reasonable predictor within the benchmark for which they were chosen and an unreliable one outside it. The residualised probe is not proposed as a better predictor—it is deliberately handicapped—but as the measurement showing that the part of the signal which survives the controls is also the part that generalises.

For the adaptive computation discussed in Chapter 2, this finding identifies a potential source of information for deciding how to approach a question. A difficulty score available during question processing could inform whether to decompose it or how much reasoning to allow. Turning that signal into a decision rule calls for calibration against answer quality and execution cost.

Once a decision to decompose has been made, the next questions concern the plan itself: does it retrieve the evidence its subquestions need, and how should a limited passage budget be distributed? Chapter 4 develops measurements for these questions, and Chapter 5 uses them to study prompt revision and passage allocation. The thesis thus progresses from anticipating compositional demand to evaluating and improving the execution of a decomposition.

3.5.2 Limitations and Future Directions

Several limitations bound the conclusions.

Difficulty is partly surface form. The surface-only probe reaches 0.91 on 2Wiki, so visible features carry real information about the benchmark labels. Residualisation removes six specified linear surface effects, not every lexical, syntactic, or semantic correlate. One it does not touch at all: 2WikiMultiHopQA is generated from templates, so its easy and hard classes may differ in template as well as in evidence-chain length, and 2Wiki is where the surface-only probe is strongest. A subsample matched on template or length would isolate this factor.

Benchmark labels are heterogeneous proxies. Hop count approximates the number of compositional steps, evidence-chain length defines a different structural split, and HotpotQA difficulty is a human annotation. HotpotQA is both the least decodable label and the source of the largest residual retention ratios because its within-benchmark reference is weak. The results decode these operational labels rather than difficulty in the abstract.

The evidence is correlational. No activation intervention or ablation tests whether the direction has a causal role; establishing one would require modifying the decoded component and measuring the resulting change in model behaviour.

The empirical scope is narrow. The study covers three open instruction-tuned models in the 24–32 billion parameter range, English Wikipedia-based multi-hop question answering, last-token pooling, and one regularisation-free linear probe family. Generalisation beyond that scope—to other model sizes, base models, architectures, languages, domains, task types, token positions, pooling choices, or nonlinear probes—is untested.

Each of these bounds indicates its own continuation: causal intervention on the decoded direction, its relation to whether a model actually executes the required reasoning, further model families and scales, and a calibrated in-model difficulty estimate for adaptive computation.

Conclusion

This chapter showed that benchmark-defined compositional question difficulty is linearly decodable from the question representations of three language models. The signal emerges across depth, survives a permutation null and partial control for surface form, behaves ordinally, and transfers across benchmarks more consistently after surface residualisation. Together, these findings identify a graded, benchmark-general component of compositional difficulty in question representations.

Having examined how compositional demand is reflected before answering, the thesis turns to what happens once a decision to decompose has been made. The challenge is then to construct useful subquestions, retrieve their supporting evidence, and distribute the available passage budget effectively. Chapter 4 establishes how to evaluate these requirements; Chapter 5 studies how to improve them.

Chapter 4
Evidence-Aware Evaluation of Question Decomposition

Introduction

Once a decision to decompose a question has been made, the challenge is to turn its subquestions into useful retrieval and reading steps. Evaluating that process requires knowing where the annotated evidence appears and estimating which subquestions are responsible for retrieving it. Following Chapter 3's study of compositional difficulty,this chapter develops an evidence-aware framework for evaluating the execution of a decomposition.

The framework combines a matrix of evidence ranks with a learned ownership model. It measures complete evidence availability, the depth required by assigned subquestions, and the answers produced during execution. We validate the ownership predictor on reference-style decompositions and compare direct retrieval with generated decomposition on 100 MuSiQue questions. A second analysis examines how a fixed passage budget can be distributed across the retrieval lists, including BM25 experiments on three benchmarks.

These studies establish three foundations for Chapter 5. The ownership-masked score supplies an objective for prompt optimisation. Balanced and oracle allocations quantify the opportunity for learned passage selection. The distinction between evidence availability, assignment, and answer execution provides the structure for diagnostic feedback. The chapter develops these foundations in turn, from measurement and validation to retrieval results and budget analysis.

4.1 Problem Formulation

4.1.1 Sequential Execution of a Decomposition

Let q be a question with reference answer y and annotated supporting passages $E(q) = \{e_1, \ldots, e_h\}$. MuSiQue supplies both supporting passages and a reference decomposition into connected single-hop questions [4]. A generated decomposition is an ordered sequence $U(q) = (u_1, \ldots, u_A)$. An item u_a may contain a reference to the answer of an earlier item. The execution substitutes the predicted earlier answer, retrieves passages for the resulting query $\widetilde{u}_a$ and produces an intermediate answer $\widehat{y}_a$:

$$\widetilde{u}_a = \text{substitute}(u_a, \widehat{y}_1, \ldots, \widehat{y}_{a-1}), \qquad \widehat{y}_a = \text{reader}\big(\widetilde{u}_a, R_a[1:k]\big). \tag{4.1}$$

Here $R_a = (d_{a1}, d_{a2}, \ldots)$ is the ranked passage list returned for $\widetilde{u}_a$, and k is the number of passages supplied to the reader. Backward references make the execution sequential even when the decomposition contains conceptually independent branches. The retained implementation returns $\widehat{y}_A$ as the final answer. Each step receives its own retrieved context; the final step has no additional synthesis call over the union of all earlier passages. One such step, a resolved query with its retrieval and its reader call, is an arm of the execution.

The number of generated subquestions A may differ from the reference hop count h. One subquestion can combine several reference information needs. Several subquestions can address the same need. The evaluation therefore permits many-to-many assignments instead of requiring an alignment by position.

4.1.2 Evaluation Requirements

A useful decomposition must make the supporting evidence available, direct it to the subquestions that need it, and produce the requested answer. We distinguish these requirements as follows.

Availability. Every annotated supporting passage occurs in at least one retrieved prefix.

Assignment. Every supporting passage has an identified owner, and each owning subquestion retrieves the passages assigned to it.

Execution. The reader produces the required intermediate answers and the final answer matches the reference under the specified answer metric.

These properties answer different questions. Evidence retrieved for a later subquestion may have been unavailable when an earlier answer was produced. Conversely, a model may answer correctly from a passage outside the annotated evidence set or from parametric knowledge. The measurements below concern recovery of benchmark evidence and reference-answer accuracy. Their relation is examined empirically rather than imposed by definition.

4.2 Evidence-Aware Retrieval Measurements

We represent an execution through the ranks of its supporting passages. This representation supports two complementary measurements: the depth at which all evidence becomes available across the lists, and the depth at which each subquestion retrieves its assigned evidence. Their aggregate scores allow decomposition instructions to be compared on a common basis.

4.2.1 Evidence Ranks and Union Depth

The starting point is the rank of each supporting passage in each subquestion's retrieval list. For a recorded execution, define an $A \times h$ matrix

$$\rho_{ai} = \min\{r : d_{ar} = e_i\}. \tag{4.2}$$

The evaluation retains the first $K = 200$ ranked passages per subquestion. When e_i is absent from that prefix, its recorded rank is $+\infty$. This value denotes censoring at K,

including evidence that might occur deeper in the full retrieval list. Passage matching uses the benchmark passage identity, determined from its title and text.

Each row corresponds to a subquestion and each column to a supporting passage. The matrix first allows us to measure evidence availability without assuming which subquestion should retrieve each passage. The union depth is the smallest common prefix length that makes all annotated evidence available somewhere in the retrieval lists:

$$D_{\text{union}}(q) = \max_{1 \le i \le h} \min_{1 \le a \le A} \rho_{ai}. \tag{4.3}$$

Consequently, $D_{\text{union}}(q) \le k$ exactly when $E(q) \subseteq \bigcup_a R_a[1:k]$. The inner minimum allows one successful retriever to satisfy the availability requirement for a passage. The outer maximum requires every annotated passage to be present.

This definition is appropriate for measuring a pooled evidence collection. In the sequential execution of Equation 4.1, however, a passage can appear in the pooled collection without reaching the particular reader call that needed it. Union depth is therefore reported alongside an assignment-aware measurement.

4.2.2 Ownership-Masked Depth

To account for the evidence needed by each subquestion, let $M \in \{0,1\}^{A \times h}$ be an ownership mask, with $M_{ai} = 1$ when subquestion a is assigned responsibility for passage e_i. The ownership-masked depth is the smallest common depth at which every assigned subquestion–passage pair is retrieved:

$$D_M(q) = \begin{cases} \max\limits_{(a,i):M_{ai}=1} \rho_{ai}, & \text{if every column of } M \text{ contains an owner,} \\ +\infty, & \text{otherwise.} \end{cases} \tag{4.4}$$

An assigned cell whose evidence is absent from the retained list also makes $D_M(q) = +\infty$. The evaluation distinguishes this retrieval censoring from an unowned column, which indicates an uncovered assignment under the predicted mask. When M is the benchmark reference assignment, D_M is computed from known ownership labels. For generated decompositions, M is supplied by the learned estimator introduced in Section 4.3; in that setting, D_M should be interpreted as an estimated ownership-masked depth, rather than a ground-truth assignment measurement.

The row and column structure has a direct interpretation. A zero row is an idle subquestion with respect to the annotated evidence. A row with several ones assigns several evidence items to one subquestion. A zero column leaves an evidence item unowned. A column with several ones assigns that evidence to several subquestions. Idle rows contribute no retrieval obligation to Equation 4.4; their generation and reading costs remain part of the executed system.

When a passage has several owners, each owner must retrieve it. This requirement reflects the separate evidence needs of the reader calls. Allowing any one owner to satisfy the others would instead measure pooled availability.

For comparison, the all-cells maximum is

$$D_{\text{all}}(q) = \max_{a,i} \rho_{ai}. \tag{4.5}$$

It requires every subquestion to retrieve every supporting passage. Whenever the mask assigns every evidence item, the three quantities satisfy

$$D_{\text{union}}(q) \le D_M(q) \le D_{\text{all}}(q). \tag{4.6}$$

The first inequality follows because each selected rank is at least the smallest rank in its column. The second follows because the masked maximum ranges over a subset of all cells. A zero column is handled separately by the censoring rule in Equation 4.4.

For example, consider

$$\rho = \begin{pmatrix} 1 & 80 \\ 40 & 2 \end{pmatrix}, \qquad M = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{4.7}$$

Both union depth and ownership-masked depth equal 2, while the all-cells maximum equals 80. Each subquestion retrieves its own assigned passage near the beginning of its list. Requiring the off-diagonal passages changes the question being evaluated. With a single full-question query, the control assigns all evidence to that query, so the three depth definitions coincide.

The distinction becomes visible if the two diagonal ranks in this example are 40 and 80, and both off-diagonal ranks are 1. Union depth is then 1, because both passages are found somewhere, but masked depth is 80: the intended owners retrieve their evidence much later.

4.2.3 Complete Coverage and the Depth Score

For a panel of n questions, complete-evidence coverage at depth k is

$$C_D(k) = \frac{1}{n} \sum_{j=1}^{n} \mathbf{1}\{D(q_j) \leq k\}, \tag{4.8}$$

where D is either D_{union} or D_M. Coverage is a question-level complete-set measurement. A question missing one required passage receives the same coverage value as a question missing several, while the retained rank matrix identifies the individual missing items.

Prompt optimisation also requires a scalar objective. The retained objective assigns logarithmic credit to finite depths:

$$v_K(d) = \begin{cases} 1 - \dfrac{\log d}{\log K}, & 1 \leq d \leq K, \\ 0, & d = +\infty, \end{cases} \qquad J_D = \frac{100}{n} \sum_{j=1}^{n} v_K(D(q_j)). \tag{4.9}$$

The score is 100 when all required evidence appears at rank 1 and 0 when every question is censored or reaches depth K. Questions with unowned evidence remain in the denominator. Reporting $C_D(K)$ beside J_D distinguishes incomplete retrieval from complete retrieval at the search limit. The score summarises complete coverage and depth. Arm count and model tokens are recorded separately because Equation 4.9 does not price them.

The reader's five-passage context and the saved 200-passage ranking serve different purposes. The former determines the evidence available during execution; the latter reveals how far below that context missing evidence lies. For example, reducing masked depth from 80 to 10 improves the score without necessarily changing the top-five contexts. The score therefore measures retrieval progress, while answer metrics assess its effect on the executed reader.

4.3 Learning Evidence Ownership

Generated subquestions can split, combine, or share the information needs of a reference decomposition. To assign evidence without assuming a positional alignment, we develop a learned ownership estimator using labelled subquestion–passage pairs. Its predictions supply the mask used by the depth measurement. The following subsections describe its construction, followed by validation in Section 4.5.

4.3.1 Training Data and Reference Assignments

MuSiQue associates each reference subquestion with a supporting passage. Substituting the reference intermediate answers resolves its dependencies and provides labelled subquestion–passage pairs. The associated passage is a positive ownership example; the other supporting passages of the same original question provide negative examples. These labels encode the benchmark's reference assignment. An alternative valid decomposition can have a different assignment.

Training uses a curated set of 6 847 MuSiQue training questions. The resulting training matrix contains 36 694 cells, of which 15 452 are positive. The held-out control contains 100 questions with reference-style subquestions and known reference assignments: 48 two-hop, 30 three-hop, and 22 four-hop questions. Training and validation therefore cover several reference reasoning depths.

The held-out question identifiers and exact resolved-subquestion–passage pairs are absent from the training set. Within the training set, questions sharing an exact repeated pair are assigned together to prevent that pair from appearing in both fitting and evaluation folds.

4.3.2 Features for Ownership Prediction

For each candidate cell, a 193-dimensional feature vector describes the relation between the resolved subquestion and the supporting passage. The features combine passage-level and sentence-level ColBERT scores, lexical similarity, token alignment, title overlap and features of the subquestion before intermediate-answer substitution. ColBERT supplies contextual token representations with a late interaction score [9].

Absolute support and relative support are both represented. Background normalisation compares semantic scores against a fixed pool of 300 passages drawn from training data. Row features compare one subquestion's support for different evidence passages. Column features compare different subquestions' support for the same passage. Agreement features record whether different similarity measures identify the same candidate.

These comparisons are relevant to ownership: two passages may share an entity, while only one answers the relation requested by the subquestion. They also permit a cell to receive low support even when it is the best option in its row. The predictor can consequently leave a row or column empty. Retrieval rank, measured depth, reference hop position and the diagonal reference label are excluded from the feature vector. The predictor is fitted to textual assignment labels rather than to the retrieval outcomes that it will help evaluate.

The estimator takes the annotated supporting passages as its candidate set and predicts their assignment to subquestions. It therefore serves as an offline evaluation component on questions for which supporting evidence is already annotated; discovering the

relevant evidence set for a new question is outside its role.

4.3.3 Training, Threshold Selection, and Ensemble Prediction

Five grouped folds are constructed while balancing reference hop cohorts. Each rotation uses three folds to fit a weighted logistic model, one disjoint fold to select its threshold and one fold for evaluation. Regularisation is selected within the fitting portion. Standardisation parameters are estimated from the same fitting data. The retained configuration gives positive ownership examples weight 3.

For model b, let $p_b(a, i)$ be its predicted ownership score. Its threshold is obtained from the lower tail of scores assigned to positive calibration cells. If those scores in ascending order are $p_{(1)}, \ldots, p_{(m)}$, the rule is

$$t_b = p_{(\lfloor (m+1)\alpha \rfloor)}, \qquad \alpha = 0.01, \tag{4.10}$$

with threshold $-\infty$ when the indicated index is zero. The final mask uses a majority vote:

$$\widehat{M}_{ai} = \mathbf{1}\left\{ \sum_{b=1}^{5} \mathbf{1}\{p_b(a, i) \geq t_b\} \geq 3 \right\}. \tag{4.11}$$

The lower-quantile threshold favours retaining true ownership assignments. Here α specifies the threshold-selection rule. Empirical recall, precision and complete-mask agreement quantify the resulting five-model procedure. Recall and precision are computed over mask cells pooled across questions, complete-mask agreement is the fraction of questions whose predicted mask matches the reference in every cell, and an arm row is exact when the prediction matches the reference in every cell of that row. These measurements assess the ensemble at the passage-pair, subquestion, and complete-plan levels.

4.4 Experimental Protocol

4.4.1 Evaluation Questions, Systems, and Execution Settings

The retrieval study uses the same 100 MuSiQue questions across the compared configurations. Its two language-model settings use GPT-4o-mini and Llama-3.3-70B-Instruct respectively for decomposition and reading. Retrieval uses HippoRAG 2 with NV-Embed-v2 embeddings and the model-specific graph and reranking configuration. HippoRAG 2 combines passage information with graph retrieval based on personalised PageRank [14]. Retrieval runs over the full MuSiQue passage corpus of 11 656 passages rather than per-question candidate sets. Each reader call receives five passages; the retained rankings extend to 200 passages for depth analysis.

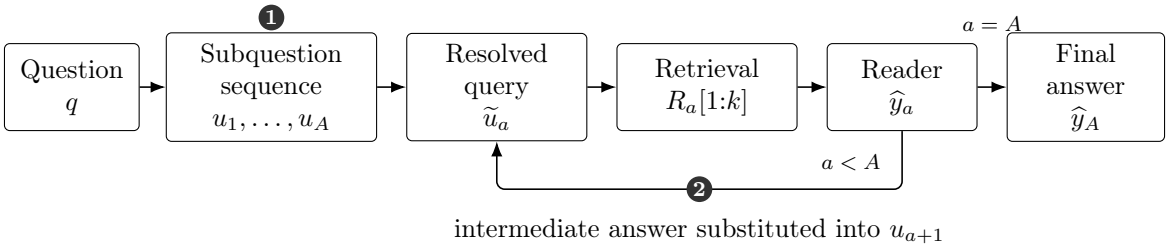
4.4.2 Compared Configurations and Evaluation Procedure

Three generated-decomposition prompts are compared with direct full-question retrieval. The baseline decomposition prompt serves as the reference generated policy, while the atomic-query prompt requests atomic single-fact queries. Both prompts use the same three demonstrations. The extended-instruction prompt uses a more detailed instruction specification without those demonstrations. These are different prompting

configurations, so their comparison measures the full prompting choice rather than instruction length alone.

Two GPT-4o-mini controls use the reference decomposition. The first substitutes gold intermediate answers into downstream queries. The second substitutes the reader's predicted intermediate answers. Both retain reference subquestions; the gold-answer condition isolates the effect of supplying correct intermediate bindings within this pipeline.

Figure 4.1 shows the execution shared by these configurations and states what each condition supplies at the two points that the comparison varies.



Condition	❶ Subquestion sequence	❷ Intermediate bindings
Direct	None: the full question is the only query	None: one retrieval and one reader call
Generated plan	Prompt-generated (baseline, atomic-query, extended-instruction)	Reader's predicted intermediate answers
Reference plan (predicted)	MuSiQue reference decomposition	Reader's predicted intermediate answers
Reference plan (gold)	MuSiQue reference decomposition	Gold intermediate answers (oracle-assisted)

Figure 4.1: Execution conditions in the retrieval study. The diagram is the execution shared by every condition, with the two marked points that the comparison varies: the source of the subquestion sequence and the source of the intermediate answers substituted into later subquestions. The table states what each condition supplies at those two points. Direct execution has neither, and the gold-binding condition is oracle-assisted.

Measurements and comparisons. Answer quality is measured by exact match (EM) and token F1 against the reference answer and its accepted aliases, under the benchmark's answer normalisation: lowercasing, removal of articles and punctuation, and collapse of whitespace [4]. Let $t(\cdot)$ denote the token multiset of a normalised string, and let $\mathcal{A}(y)$ denote the reference answer y together with its accepted aliases. Then

$$\mathrm{EM}(\widehat{y}, y) = \mathbf{1}\{\widehat{y} \in \mathcal{A}(y)\}, \qquad \mathrm{F1}(\widehat{y}, y) = \max_{y' \in \mathcal{A}(y)} \frac{2|t(\widehat{y}) \cap t(y')|}{|t(\widehat{y})| + |t(y')|}, \tag{4.12}$$

where all compared strings are normalised and $\cap$ is the multiset intersection. Both quantities are averaged over the panel; EM is reported as a percentage and F1 on a 0–100 scale.

Retrieval is measured by complete-evidence coverage (Equation 4.8), union depth (Equation 4.3) and ownership-masked depth (Equation 4.4), summarised by the depth score J_D of Equation 4.9. The number of generated arms, retrieved passage slots and recorded total model tokens describe resource use. A passage slot is one passage delivered to one reader call. Per-arm passage counts therefore charge duplicate passages again when

they are sent to separate reader calls, while unique passage counts describe the pooled collection. Recorded tokens are the prompt and completion tokens of the decomposition and reading calls of a question.

The reference-answer control records one token total across several execution branches. Per-branch costs are therefore unavailable. Generated and direct runs have separate recorded totals, which are reported per question.

Comparisons pair the same questions across configurations. Where intervals are reported, they are percentile intervals from 10 000 paired question-level bootstrap resamples. They describe variation within this development panel. Several prompt and attribution configurations were examined on the panel, so these intervals are descriptive rather than a correction for adaptive model selection. The reference-style ownership control was kept separate from weight fitting and threshold calibration; its role is a structured assignment control rather than an evaluation of arbitrary generated plans.

4.5 Ownership Validation and Decomposition Results

The experiments assess both the evaluation framework and the decompositions it measures. Ownership recovery tests whether the estimator can reproduce known evidence assignments. The retrieval and answer comparisons then compare generated decomposition with direct retrieval and examine how intermediate answers affect the resulting execution.

4.5.1 Ownership Prediction Accuracy

The ownership ensemble recovers the complete reference assignment on 92 of 100 questions, with precision of 97.46% and recall of 98.18%. Table 4.1 reports the detailed measurements. The model recovers 269 of 274 reference owner cells and selects seven non-owner cells. At the subquestion level, 264 of 274 arm rows are exact. These results support automatic ownership measurement on the evaluated reference-style plans.

Table 4.1: Ownership prediction on the reference-style decomposition control. Cell metrics use 814 subquestion–passage pairs. Complete-mask agreement is measured over the 100 questions.

Measurement	Value
True positives / false negatives	269 / 5
False positives / true negatives	7 / 533
Precision	97.46%
Recall	98.18%
Cell accuracy	98.53%
AUROC of mean model scores	0.9977
Exact arm rows	264/274
Exact question masks	92/100

The cohort breakdown in Table 4.2 shows the distribution of the twelve cell errors. Four-hop questions contain three of the five missed owners. Because several cells belong to the same question, cell totals are reported with their question counts rather than interpreted as independent question outcomes.

Table 4.2: Ownership errors by reference hop count. TP, FN, FP and TN count individual cells, while n counts questions.

Reference hops	n	Cells	TP	FN	FP	TN
2	48	192	95	1	4	92
3	30	270	89	1	0	180
4	22	352	85	3	3	261

On the gold-answer retrieval traces, the exact reference mask yields complete assigned-evidence coverage of 100% at depth 200 and a score of 86.47. Using the learned ownership mask gives 97% coverage and a score of 84.84. This comparison isolates the effect of replacing the reference assignment with its prediction on those fixed lists.

4.5.2 Sensitivity to Ownership Predictions

Ownership predictions determine which ranks enter the masked depth. Adding an owner adds a retrieval requirement and can increase the depth. Removing an owner can reduce it if another owner remains; removing the last owner of a required passage instead makes the depth infinite. Assignment errors can therefore move the score in either direction. To distinguish these changes from retrieval improvements, the following comparison holds the ranked lists fixed and varies only the ownership model.

Table 4.3 scores the same retrieval traces under two attribution models: an earlier predictor built on a different feature set, and the retained model of Section 4.3. The baseline generated decomposition receives scores of 52.32 and 72.19 respectively. The analogous atomic-query values are 52.81 and 66.64. These changes arise from the assigned cells, with the passages and ranks held fixed. On the reference arms with gold bindings the two models differ by 24.98 points, while replacing the retained prediction with the exact reference assignment changes the score by 1.63. The masked-depth measurement is therefore more sensitive to the choice of attribution model than to the residual error of the retained one.

Table 4.3: Masked-depth measurements under two attribution models on fixed GPT-4o-mini retrieval traces. Coverage is complete assigned-evidence coverage at $K = 200$; score is J_{D_M} from Equation 4.9. The passages and their ranks are identical across rows, so only the assigned cells differ. The last row uses the exact reference assignment.

Decomposition	Attribution	Coverage (%)	Score
Baseline	Earlier model	78	52.32
Baseline	Retained model	87	72.19
Atomic-query	Earlier model	77	52.81
Atomic-query	Retained model	81	66.64
Reference, gold bindings	Earlier model	88	59.86
Reference, gold bindings	Retained model	97	84.84
Reference, gold bindings	Reference assignment	100	86.47

The reference-style control supports the use of the retained predictor for recovering known assignments. For generated arms, assignments can legitimately be fused or shared. The score differences in Table 4.3 therefore establish sensitivity to attribution rather than a labelled accuracy comparison on generated plans. Absolute masked-depth values

on generated decompositions should therefore be interpreted as estimator-dependent diagnostics. Comparisons between prompts hold this source of variation fixed and are reported alongside ownership-independent union coverage and final-answer measurements.

4.5.3 Decomposition versus Direct Retrieval

For readability, $C_U(k)$ denotes the union coverage $C_{D_{\text{union}}}(k)$ in the result tables. Table 4.4 reports the direct and generated-decomposition executions. With GPT-4o-mini, the baseline decomposition raises complete union coverage at five passages per arm from 45% to 76%. EM changes from 40% to 52% and F1 from 52.64 to 59.00. With Llama-3.3-70B, complete union coverage changes from 46% to 79%, while EM changes from 42% to 43% and F1 from 52.30 to 51.37.

Table 4.4: Recorded execution results on 100 MuSiQue questions. EM and F1 are percentages. $C_U(5)$ is complete union coverage at five passages per arm. Arms and tokens are per-question means. Retrieved evidence and final answers are separate outcomes.

Model	Prompt	EM	F1	$C_U(5)$	Arms	Tokens
GPT-4o-mini	Direct	40	52.64	45	1.00	4 369
	Baseline	52	59.00	76	2.79	12 763
	Atomic-query	52	59.66	72	2.66	11 898
	Extended-instruction	50	60.39	74	2.61	13 881
Llama-3.3-70B	Direct	42	52.30	46	1.00	4 530
	Baseline	43	51.37	79	3.00	14 069
	Atomic-query	45	53.54	78	3.22	14 789
	Extended-instruction	47	55.94	75	2.84	15 271

The paired bootstrap interval for the baseline decomposition's EM change is $[2, 22]$ percentage points for GPT-4o-mini and $[-8, 10]$ for Llama-3.3-70B. On this 100-question development panel, the GPT-4o-mini decomposition is associated with higher answer accuracy as well as higher evidence coverage. This end-to-end comparison is not compute-matched, since decomposition uses additional passage slots and reader calls; Section 4.6 isolates retrieval behaviour under a matched passage-slot budget. The Llama result shows that increased evidence coverage does not necessarily translate into a comparable answer-quality gain, motivating the controlled intermediate-answer experiment below.

The per-arm comparison also changes the total retrieval expenditure. The baseline GPT-4o-mini decomposition retrieves a mean of 13.95 passage slots across its reader calls, compared with 5 for direct retrieval. Its mean pooled collection contains 11.35 distinct passages. Thus the coverage increase in Table 4.4 combines a change in query formulation with more passage slots and more reader calls. Section 4.6 compares fixed lists at a common total passage budget.

4.5.4 The Effect of Intermediate Answers

The two reference-decomposition controls use the same subquestion templates and reader but change the answers substituted into later queries. Table 4.5 reports their outcomes. Gold substitution yields EM 64% versus 49% under predicted substitution, with a paired bootstrap interval of $[8, 22]$ percentage points for the difference. Complete union coverage at five passages per arm changes from 80% to 92%.

Table 4.5: Reference-decomposition controls with GPT-4o-mini. Gold bindings supply reference intermediate answers when constructing downstream queries. Predicted bindings use the reader's earlier outputs.

Intermediate substitution	EM (%)	F1	$C_U(5)$ (%)
Predicted answers	49	57.50	80
Gold answers	64	70.73	92

The control identifies intermediate answers as a point of intervention in the retrieval chain. Correct substitutions improve downstream query text and the evidence available for answering. Because this experiment supplies gold answers, it measures the benefit of correcting intermediate state within the reference pipeline. This finding motivates recording intermediate predictions alongside retrieval outcomes when constructing optimisation feedback.

Reference and generated queries with BM25. A complementary MuSiQue comparison examines reference plans with gold intermediate answers and generated plans with predicted answers under BM25. Both search the same 11 656-passage corpus. Generated traces use five passages per intermediate reader call, and each ranked list is saved to depth 200. Table 4.6 reports complete-evidence coverage when a total budget of 8 or 200 passage slots is divided as evenly as possible across the subquestions. Section 4.6 defines this allocation formally, and Section 4.6.4 gives the BM25 panel and trace-generation details.

Table 4.6: MuSiQue BM25 retrieval regimes. Coverage (%) is measured on each retained panel under balanced slot allocation at total budgets $B = 8$ and $B = 200$. The reference row uses a different plan source and a larger panel.

Query regime	n	$B = 8$	$B = 200$
Reference plan, gold bindings	1 000	59.5	96.2
Generated plan, predicted bindings	824	41.3	73.3

The reference regime achieves higher coverage at both budgets, reaching 96.2% at $B = 200$, compared with 73.3% for generated plans. This is a descriptive, oracle-assisted comparison: both the plan source and the intermediate answers change, and the panels contain 1 000 reference questions and 824 retained generated traces. The gap therefore cannot be attributed to intermediate-answer quality alone. The controlled reference-chain experiment above isolates that change while holding the subquestion templates fixed.

4.6 Evidence Coverage Under a Passage Budget

The evidence-rank framework also makes it possible to study how passages should be distributed across subquestions. We compare a balanced allocation with a gold-informed oracle under the same total passage budget, holding the recorded queries fixed. The resulting coverage gap quantifies the opportunity for a learned allocator. We evaluate it first on the original MuSiQue panel and then with BM25 across three benchmarks.

4.6.1 Passage-Slot Accounting and Balanced Allocation

A common per-arm depth gives more total passage slots to longer decompositions. To compare questions under the same total allowance, consider a per-question ceiling $B \leq AK$ distributed among the A recorded retrieval lists. Balanced allocation uses the whole allowance and assigns

$$k_a(B) = \left\lfloor \frac{B}{A} \right\rfloor + \mathbf{1}\{a \leq B \bmod A\}, \qquad \sum_{a=1}^{A} k_a(B) = B. \tag{4.13}$$

The remainder is allocated in execution order. For example, a budget of eight slots shared by three arms gives depths $(3, 3, 2)$. Each occurrence of a passage in a selected prefix costs one slot, including duplicates across arms. Complete evidence availability is evaluated in the pooled set $\bigcup_a R_a[1 : k_a(B)]$; no ownership mask is needed for this union criterion.

This allowance counts selected passage slots. The same number of slots can produce different numbers of unique passages and different token totals, since passages can recur across arms and vary in length. Retrieving the ranked lists, producing the queries and reading the contexts incur separate computational costs.

These are fixed-list replays. The original intermediate answers and later queries were generated using five passages per reader call. Changing a prefix in the replay leaves those earlier answers and later queries unchanged. The result measures the coverage of the recorded lists at a specified slot budget. An end-to-end policy evaluated at that budget would regenerate intermediate answers and dependent queries under its actual contexts.

The ceiling applies to each question separately. Its largest feasible common depth is $\lfloor B/A(q) \rfloor$, with the balanced rule also spending the remainder. A single depth shared across the panel would be limited by the longest plan to $\lfloor B/\max_q A(q) \rfloor$, leaving slots unused for shorter plans. Allowing transfers between questions under an average budget would define a different allocation problem.

4.6.2 Oracle Allocation on Recorded Lists

Supporting passages need not occur at similar ranks across arms. For example, if two arms contain distinct required passages at ranks two and six, with neither passage available elsewhere, a balanced $(4, 4)$ allocation misses one passage whereas $(2, 6)$ covers both at the same cost. The oracle uses gold evidence identities to find the least expensive prefix allocation that covers a question:

$$B^*(q) = \min_{k_1,\ldots,k_A \in \{0,\ldots,K\}} \left\{ \sum_a k_a : E(q) \subseteq \bigcup_a R_a[1 : k_a] \right\}. \tag{4.14}$$

The value is $+\infty$ when the retained lists jointly miss annotated evidence. For finite cases, it can be computed by dynamic programming over subsets of evidence. Let $S_a(k)$ denote the evidence indices found in arm a's first k passages. Starting from $F_0(\varnothing) = 0$, the update is

$$F_a(T \cup S_a(k)) \leftarrow \min\{F_a(T \cup S_a(k)), F_{a-1}(T) + k\}. \tag{4.15}$$

Unreachable states have value $+\infty$, and $B^*(q) = F_A(\{1, \ldots, h\})$. Only zero and prefix lengths at which an additional evidence item appears need to be considered: a longer prefix with the same evidence set adds cost without changing that state. With at most four annotated hops in these experiments, the subset state space is small.

The oracle uses gold evidence identities to choose its allocation. Moreover, all recorded queries are available even when their selected prefix length is zero. The objective charges passage slots but leaves the generation of those queries and their intermediate states outside the optimisation. It is an exact minimum for the fixed-list problem and an optimistic reference for a system that must construct and execute the queries. It is not an achieved runtime or token saving.

Applying the coverage aggregation in Equation 4.8 to these minimum budgets gives the oracle coverage frontier:

$$C^*(B) = \frac{1}{n} \sum_{q=1}^{n} \mathbf{1}\{B^*(q) \leq B\}. \tag{4.16}$$

It is the fraction of questions whose complete evidence can be recovered within B slots under the best prefix allocation. A question missing a required passage from every saved list remains uncovered at every budget.

4.6.3 Allocation Results on the MuSiQue Panel

The replay uses the same 100 MuSiQue questions and HippoRAG 2 rankings as the execution study. Its budget of $B = 15$ is close to the baseline GPT-4o-mini decomposition's mean of 13.95 passage slots, giving the comparison a similar selected-passage allowance. At $B = 15$, balanced allocation recovers complete evidence for 76 questions under the baseline GPT-4o-mini decomposition and 78 under its Llama counterpart. The corresponding oracle allocations reach 85 and 87; an oracle entry counts the questions whose minimum feasible budget $B^*(q)$ in Equation 4.14 is at most 15. Table 4.7 reports the same comparison for the other prompts and for direct retrieval, which are the configurations available without a reference decomposition. Direct retrieval has a single arm, so its two allocations coincide by construction, and its entries give that one list 15 passages rather than the five used in execution.

Table 4.7: Complete evidence coverage (%) at a total budget of 15 passage slots on fixed recorded lists. The oracle chooses prefixes using gold evidence and is a reference allocation rather than a deployed policy. Each row contains 100 questions.

Model	Prompt	Balanced allocation	Oracle allocation
GPT-4o-mini	Direct	66	66
	Baseline	76	85
	Atomic-query	73	85
	Extended-instruction	75	83
Llama-3.3-70B	Direct	65	65
	Baseline	78	87
	Atomic-query	75	88
	Extended-instruction	75	85

On these fixed recorded rankings, the baseline decompositions retain a complete-evidence coverage advantage over direct retrieval after matching the total selected passage-slot budget. The additional nine questions recoverable by oracle allocation on the baseline decomposition in each model setting show that redistributing slots can recover evidence missed by the balanced split. This provides a concrete target for the learned policies in

Chapter 5: recover more supporting evidence by choosing depths from observable query and retrieval features.

4.6.4 Cross-Benchmark BM25 Evaluation

The next study examines whether the allocation opportunity also appears with a different retriever and on other datasets. It uses BM25 over separate closed corpora containing 11 656 passages for MuSiQue, 10 426 for 2WikiMultiHopQA [3], and 9 811 for HotpotQA [2]. The MuSiQue corpus is unchanged from the earlier experiment; the question panel is larger.

Evaluation population and trace generation. Each source panel contains 1 000 questions. Within the available compute budget, trace generation retained 824 MuSiQue, 977 2WikiMultiHopQA, and 996 HotpotQA questions. All allocation comparisons and the full-question control use the same retained IDs within each benchmark. Retention follows execution order and per-question cost, not retrieval or answer outcomes. Coverage therefore describes the retained questions. These panels also informed development and are separate from the frozen comparison in Chapter 5.

Generated subquestions are executed sequentially with GPT-4o-mini. Each retrieval call searches the full benchmark corpus, the reader receives the top five passages, and its predicted answer resolves references in later subquestions. The top 200 passages from each ranked list are saved. Allocation uses these fixed lists and the passage-slot accounting defined above, preserving the answers and queries produced during trace generation.

Coverage at a matched budget. Table 4.8 compares full-question retrieval, balanced allocation, and the oracle at $B = 8$. Full-question retrieval selects eight passages from its single list. Balanced allocation divides the same budget among the generated subquestions, while the oracle chooses their prefix lengths using gold evidence ranks. The recorded plans contain at most seven arms on MuSiQue and HotpotQA and six on 2Wiki, so balanced allocation can give every arm at least one slot.

Table 4.8: Complete annotated-evidence coverage (%) at a ceiling of eight selected passage slots. BM25 results use fixed predicted-binding traces; the oracle uses gold evidence ranks. Counts are retained questions from each original 1,000-question panel.

Dataset	n	Full question	Balanced	Oracle
MuSiQue	824	13.6	41.3	53.6
2Wiki	977	31.5	68.8	88.1
HotpotQA	996	63.7	76.2	83.6

On the retained fixed traces, balanced allocation achieves higher complete annotated-evidence coverage than full-question retrieval at $B = 8$ on all three benchmarks, while the gold-informed oracle provides additional headroom. On MuSiQue, for example, coverage rises from 13.6% with full-question retrieval to 41.3% with balanced allocation and 53.6% with the oracle. The gap between balanced and oracle allocation shows the opportunity to improve coverage by distributing the same passage budget unevenly across arms. Some lists require deeper prefixes to reach supporting evidence, while others can contribute their evidence with fewer slots. This allocation headroom therefore also appears with BM25 and across all three benchmarks.

The dynamic program in Equation 4.15 independently reproduces the saved oracle coverage on all three panels at the checked budgets.

Budgets needed for target coverage. The same comparison can be read in terms of the budget needed to reach a coverage target. On the MuSiQue sequential panel, balanced allocation first reaches 50%, 60%, and 70% coverage at 16, 32, and 128 slots, respectively. The oracle reaches these targets at 8, 16, and 48 slots. These are the first tested budgets that meet each target, rather than exact minimum budgets over all possible values.

At $B = 200$, the MuSiQue oracle reaches 79.5% coverage, compared with 73.3% under balanced allocation. This comparison uses the generated regime from Table 4.6. The remaining oracle gap concerns redistribution within the saved lists. Improving the queries or changing the intermediate reader contexts would require new execution; neither effect is measured by this replay.

Finally, allocation depends on the number of generated arms, whose means are 2.53, 2.79, and 2.10 on MuSiQue, 2Wiki, and HotpotQA, respectively. More arms provide more allocation choices, but may also reflect redundant splitting or prompt preferences. Any association between arm count and allocation headroom therefore concerns the generated plan structure; it does not by itself show that harder questions offer greater savings.

4.7 Discussion and Implications for Optimisation

4.7.1 Contributions and Implications

The framework links evidence responsibility to retrieval depth. Ownership assignments distinguish pooled evidence availability from retrieval by the intended subquestion, complementing final-answer measurements.

On the 100-question MuSiQue development panel, generated decomposition increases evidence coverage under both reader settings and is associated with higher final-answer accuracy for GPT-4o-mini. Because the end-to-end comparison uses more passage slots and reader calls, the separate fixed-list replay evaluates retrieval under matched passage-slot budgets. Matched-budget replay shows that redistributing passage slots can recover evidence missed by balanced allocation, an opportunity also observed with BM25 across three benchmarks.

Chapter 5 applies these foundations: the ownership-masked score evaluates OPRO candidates, the oracle frontier motivates learned allocation, and the execution diagnostics structure feedback for prompt revision.

4.7.2 Scope and Limitations

Ownership validation uses reference-style decompositions with known supporting passages. Its archived template history is only partially documented, and accuracy on naturally generated plans requires independently labelled assignments. The estimator assigns a supplied evidence set; discovering that set for a new question is a separate task. Its threshold and voting procedure are assessed empirically, with simultaneous assignment guarantees requiring additional assumptions.

The retrieval comparisons describe development panels under the reported models and corpus scope. Bootstrap intervals measure variation within those panels. The reference-versus-generated BM25 comparison changes both the plans and intermediate answers on

different populations, so the controlled reference-chain experiment provides the narrower evidence about substitution. Per-branch token costs are also unavailable for the reference-answer controls.

Allocation replay preserves queries generated with five passages per intermediate reader call. Its budget counts selected slots, including repeated passages, and excludes trace generation. Oracle improvements therefore quantify the potential of the saved rankings. Measuring operational savings requires a policy evaluated with its actual reader contexts, answer quality, and execution cost.

Conclusion

This chapter connected retrieval ranks to learned evidence ownership, recovering 92 of 100 complete reference-style masks. On the MuSiQue panel, decomposition improved evidence retrieval and raised GPT-4o-mini exact match from 40% to 52%. Intermediate-answer controls further showed why plan execution matters alongside evidence availability.

Gold-informed oracle allocation further showed, across three BM25 benchmarks, that unequal allocation can recover supporting evidence missed by a balanced split, establishing the headroom studied by the learned allocator in Chapter 5. Chapter 5 turns these measurements and diagnostics into tools for prompt search and passage selection.

Chapter 5
Optimising Question Decomposition and Evidence Allocation

Introduction

Chapter 4 established three foundations for improving question decomposition: an ownership-masked retrieval score, an oracle analysis of passage allocation, and a distinction between evidence availability, assignment, and execution. This chapter develops their applications through automated prompt search, learned passage allocation, and structured evidence feedback.

The first contribution is an executable prompt-search pipeline that uses the ownership-masked score to evaluate decomposition instructions. An OPRO-style search compares 25 instructions on 100 MuSiQue questions, allowing us to study how instruction changes affect retrieval depth, answers, and cost. The second is a learned allocator that predicts depth requirements from query features and retrieval scores. Experiments on three BM25 benchmarks measure its effect on evidence coverage across several passage budgets, and a paired 700-question MuSiQue study tests the final answers produced from its selected contexts.

The third contribution is an implemented feedback representation that organises evidence ranks, ownership estimates, and intermediate predictions for prompt revision. Controlled plan transformations test its ownership diagnostics, and a separate evaluation protocol defines how to assess its future effect on search performance. The chapter presents each method with its available evidence, then draws the implications for retrieval-efficient decomposition. Detailed supplementary analyses and the prospective protocol are retained in Appendix A.

5.1 Optimisation Framework

Chapter 4 identified two opportunities for improving decomposition: changing the queries that retrieve evidence and changing how many passages are selected from each query. This chapter studies these interventions separately. Prompt search changes the decomposer's instruction and executes the resulting plans; learned allocation selects passages from fixed recorded rankings. Both are assessed through retrieval quality, final answers, and resource use.

These experiments answer different questions and use different evaluation populations. Table 5.1 provides a map of the studies across Chapters 3–5; the sections below specify

their models and protocols.

Table 5.1: Study populations and experimental roles across Chapters 3–5. Counts denote questions unless stated otherwise. All rows report completed work except the prospective feedback comparison.

Study	Population	Role and relation to other studies
Difficulty probing (Ch. 3)	600 2Wiki; 900 HotpotQA; 900 MuSiQue	Class-balanced panels for cross-validated probing and transfer analyses.
Ownership prediction (Ch. 4)	6,847 MuSiQue training questions; 100 held-out reference-style questions	Fits the ownership estimator and validates known evidence assignments.
Retrieval and execution (Ch. 4)	100 MuSiQue	Paired decomposition comparisons and intermediate-answer controls.
OPRO search (Ch. 5)	The same 100 MuSiQue questions	Development panel used for feedback, candidate selection, and reported outcomes.
BM25 allocation (Chs. 4–5)	824 MuSiQue; 977 2Wiki; 996 HotpotQA	Retained from 1,000 questions per benchmark; fixed-list baselines and question-level cross-validation of learned allocation.
Final-reader allocation (Ch. 5)	700 retained MuSiQue questions	Subset of the allocation panel; allocator refitted within this subset for paired reader evaluation.
Ownership stress tests (Ch. 5)	52 variants of 12 MuSiQue questions	Controlled plan changes testing the ownership diagnostics.
Feedback comparison (prospective)	96 optimisation; 48 selection; 200 confirmation questions	Frozen MuSiQue splits for evaluating feedback effects on search.

5.1.1 The Instruction as an Optimisation Variable

The prompt-search experiment changes the decomposer's system instruction p, while keeping the demonstrations, retriever, reader, and ownership estimator fixed. For an input question q, the decomposer produces one ordered plan

$$\pi_p(q) = (u_1, \ldots, u_{A_p(q)}), \tag{5.1}$$

where $A_p(q)$ is the number of subquestions. Execution follows the procedure in Chapter 4: each arm retrieves and reads passages, and its predicted answer resolves references in later arms. A reference `#i` in u_a must therefore satisfy $i < a$. The last arm's answer is the final output $\widehat{y}_p(q)$.

Useful instructions must produce retrieval questions with executable dependencies and a final arm that answers the original question. A plan can retrieve relevant facts yet stop at an intermediate entity, leaving the requested answer unresolved. Changing p can consequently affect query quality, intermediate answers, and the number of retrieval and reader calls.

5.1.2 The Retrieval Objective

The prompt search uses Chapter 4's ownership-masked depth score. For generated plans, the search uses the estimated ownership-masked depth supplied by the frozen ownership estimator. Let $D_{M,p}(q)$ denote this quantity for the plan generated under p: the greatest rank among assigned arm–evidence pairs, or infinity when an ownership or retained-list gap prevents complete retrieval. Written as a function of the instruction and evaluation panel $\mathcal{D}$, the objective at cutoff $K = 200$ is

$$J_M(p;\mathcal{D}) = \frac{100}{|\mathcal{D}|}\sum_{q\in\mathcal{D}} v_K(D_{M,p}(q))\,, \qquad v_K(d) = \begin{cases} 1 - \dfrac{\log d}{\log K}, & 1 \le d \le K, \\ 0, & d = +\infty. \end{cases} \tag{5.2}$$

All questions remain in the denominator. The objective rewards early complete retrieval under the predicted assignment and gives zero credit to a censored question. Its companion coverage statistic is

$$C_M(K;p) = \frac{1}{|\mathcal{D}|}\sum_{q\in\mathcal{D}} \mathbf{1}\{D_{M,p}(q) \le K\}. \tag{5.3}$$

Depth K receives zero logarithmic credit but counts as finite coverage. Reporting both quantities distinguishes complete evidence near the search limit from incomplete evidence.

The ownership estimator is frozen during search so that every candidate is evaluated by the same assignment procedure. Gold supporting passages are used only for offline development and evaluation; they are not supplied to the decomposer or reader during execution.

5.1.3 Answer Quality and Resource Use

The end-to-end outcomes are exact match and token F1 against the benchmark reference answers. For a fixed evaluation panel, define

$$J_{\mathrm{F1}}(p) = \frac{1}{|\mathcal{D}|}\sum_{q\in\mathcal{D}} \mathrm{F1}(\widehat{y}_p(q), y_q). \tag{5.4}$$

The retrieval objective in Equation 5.2 and the answer objective in Equation 5.4 measure different properties. Retrieved evidence must still be read, intermediate answers must bind correctly, and the last arm must produce the requested answer.

The prompt-search reader consumes five passages per arm. Its nominal passage-slot allowance is consequently $5A_p(q)$, with repeated passages charged to each arm that reads them. Unique passages, reader tokens and model calls are additional cost measurements. Reporting them alongside the masked-depth score shows how a prompt changes the total execution, since the score itself contains no arm-count penalty.

5.2 Automated Search over Decomposition Instructions

We integrate the ownership-masked evaluation into an automated instruction search. Each proposed prompt generates a decomposition, the pipeline executes its retrieval and reading steps, and the resulting evidence score determines selection. This turns Chapter 4's measurement into an optimisation objective. The study evaluates the selected instruction on retrieval depth, complete coverage, final answers, and passage use.

5.2.1 Search Configuration

OPRO uses a language model to propose solutions from a history of candidates and their measured scores [33]. Here, a candidate is the text of the decomposer's system instruction. GPT-4o proposes instructions, and the Llama-3.3-70B pipeline executes them on the development questions. Table 5.2 records the retained configuration.

Table 5.2: Configuration of the completed prompt-search pilot.

Component	Setting
Evaluation panel	The same 100 MuSiQue questions for every candidate
Initialisations	One initial instruction
Search rounds	Four, with six new instructions per round
Candidate total	25, including the initial instruction
Proposal model	GPT-4o, temperature 1.0
Execution model	Llama-3.3-70B, one plan per question
Reader context	Five passages per executed arm
Retrieval scoring	Up to 200 retained ranks per arm
Search reward	Ownership-masked logarithmic depth score
History in feedback	At most 20 prior instructions and their rounded scores
Example feedback	Three cases from the current leading instruction

The six instructions proposed in a round are evaluated as separate candidates. Each candidate produces one plan per question; their plans are not combined during execution. The executor uses temperature-zero generation with cached model calls. Exact duplicate instruction hashes are filtered. The retained leaderboard contains twenty-five successfully evaluated candidates and zero rejected candidates. This amounts to 2 500 candidate–question evaluations. That count describes logical evaluations; cached calls can reduce the number of billable model requests.

The baseline is the seed instruction and its executions recorded within this search. The separately recorded prompt variants in Chapter 4 provide context for the pipeline, rather than substituting for this paired baseline.

5.2.2 Search Procedure and Feedback

The search begins by executing the initial instruction on all hundred questions. At each round, the optimizer receives the retained instruction history, the corresponding aggregate depth scores and three examples from the current leader. It proposes six replacement instructions. Each is executed through the same retrieval and reading pipeline, evaluated with an estimated ownership mask from the frozen estimator, and scored using Equation 5.2. The updated leaderboard retains the highest-scoring instruction across all completed rounds.

The full-precision score determines the leaderboard order, while the meta-prompt displays integer-rounded scores. Twelve of the twenty-five candidates are displayed in the same score-64 bucket. This reduces the resolution of the numerical information available to the proposer, even though their stored scores remain distinct.

The initial and selected instructions share the same planning policy: use complete natural-language questions, preserve independent branches, require backward references and split compound lookups into simpler arms. The selected instruction adds an explicit

check that the necessary facts of the original question are covered. These are observed differences in instruction text. The experiment changes several clauses together, so the measured score change cannot be assigned to that coverage clause alone.

Examples supplied to the optimiser. The pilot's feedback prioritises questions with predicted unowned evidence, then questions with the greatest masked depth. It selects the unowned cases in stored question order. This gives the optimizer examples of poorly scored executions, but repeatedly drawing from the current leader and a fixed ordering can concentrate attention on a small part of the panel.

The example description attributes failures to facts that the plan failed to ask for. That interpretation is stronger than the underlying measurements. A passage may be assigned to an appropriate query yet appear at a large retrieval rank. Conversely, an unowned passage is an estimator output whose correctness depends on the ownership model. Both cases can lower the depth score, while suggesting different changes to the instruction. This motivates the typed feedback in Section 5.4.

5.2.3 Progress Across Search Rounds

Table 5.3 separates the best proposal in each round from the best instruction retained so far. The overall winner appears in the second round. The third and fourth rounds produce lower-scoring candidates, while the stored best score remains 67.20.

Table 5.3: Masked-depth score during the completed search. The retained best includes all earlier rounds. Values are on the shared development panel.

Stage	New candidates	Best in stage	Retained best
Initial instruction	1	66.3017	66.3017
Round 1	6	66.5975	66.5975
Round 2	6	67.2042	67.2042
Round 3	6	65.2557	67.2042
Round 4	6	66.0482	67.2042

The search selects an instruction 0.9025 score points above the initial instruction. The highest development-panel score is reached in round two and retained through the remaining rounds. This confirms the operation of the proposal, execution, scoring, and selection loop, without by itself establishing out-of-sample improvement.

5.2.4 Retrieval, Answers, and Cost of the Selected Instruction

The development-selected instruction achieves an aggregate masked-depth score 0.90 points above the seed instruction on the same 100-question panel, while finite masked coverage remains 80%. Table 5.4 compares the seed and selected instructions on retrieval, final answers, and passage use. The score gain therefore occurs with the same number of questions achieving complete assigned-evidence retrieval, although their identities can differ.

Table 5.4: Initial and selected instructions on the 100 development questions. F1 is displayed on a 0–100 scale. Passage slots are the five-per-arm allowance derived from the observed arm count.

Measurement	Initial	Selected	Change
Masked-depth score	66.3017	67.2042	+0.9025
Finite masked coverage (%)	80	80	0
Exact match (%)	44	44	0
Token F1	52.3682	52.4847	+0.1165
Mean generated arms	3.00	3.13	+0.13
Mean passage-slot allowance	15.00	15.65	+0.65

Exact match is unchanged at 44%: two questions change from incorrect to correct and two change in the opposite direction. Mean F1 increases by 0.1165 points on the 0–100 scale. The selected instruction generates 0.13 additional arms per question, corresponding to a 4.3% increase in the mean passage-slot allowance. The search changes retrieval depth and plan structure while producing similar aggregate answer quality.

A paired bootstrap over the hundred questions gives a descriptive 95% interval of $[-1.80, 3.90]$ points for the masked-depth-score difference and $[-3.60, 4.00]$ points for the F1 difference. The exact-match difference has an interval of $[-4, 4]$ percentage points. Because this panel was used throughout candidate proposal, feedback, and selection, these paired intervals do not constitute an independent test of prompt improvement. The result should be interpreted as development-set optimisation. These descriptive intervals condition on the selected pair and the development panel; their implications for generalisation are discussed in Section 5.5.2.

5.2.5 Choosing between Retrieval and Answer Objectives

The search also makes it possible to compare how retrieval and answer objectives rank the same candidate instructions. Across the twenty-five candidates, the Spearman correlation between masked-depth score and mean answer F1 is 0.413. This describes the relationship within the adaptively generated candidate set on its shared evaluation panel.

The highest-F1 candidate occurs in the third round. It achieves F1 53.88 and exact match 46%, with masked-depth score 64.10. The depth-selected winner has F1 52.48 and exact match 44%, with score 67.20. Table 5.5 shows the resulting reversal. The highest-F1 candidate is a retrospective comparator, not a second independently selected test winner.

Table 5.5: Different objectives select different candidates from the same development search. The F1-leading candidate is identified retrospectively.

Instruction	Depth score	F1	Exact match (%)
Initial	66.30	52.37	44
Highest depth score	67.20	52.48	44
Highest answer F1	64.10	53.88	46

This comparison identifies objective choice as a practical part of prompt optimisation. The masked-depth score selects for earlier retrieval of assigned evidence, while answer F1 selects for the outcome of the complete execution. It also motivates the feedback extension below: use final-answer quality for selection and retain evidence measurements to explain where a candidate's execution can be improved.

The completed search thus contributes an operational use of the depth score and an empirical comparison of selection objectives. The development-selected instruction achieves a modestly higher retrieval-oriented score while exact match remains unchanged. More importantly, candidate ranking differs across retrieval and answer objectives, showing that retrieval progress alone is not a sufficient proxy for end-to-end QA quality.

5.3 Learning to Allocate Retrieved Passages

Chapter 4's oracle analysis showed that redistributing passage slots can recover evidence missed by a balanced split. We develop a learned allocator that predicts depth requirements from observable query features and BM25 scores, then uses these predictions to distribute a fixed passage budget. The study tests whether this captures part of the oracle opportunity through coverage measurements on three benchmarks and a final-reader experiment on MuSiQue.

5.3.1 Allocation Setting and Comparators

Prompt optimisation changes a plan and its resulting rankings. Allocation holds those rankings fixed and chooses how much evidence to retain from each arm. For a question with A recorded arms, let k_a be the prefix length selected from arm a. Its passage-slot budget is

$$\sum_{a=1}^{A} k_a \leq B. \tag{5.5}$$

Complete coverage means that every annotated supporting passage occurs in at least one selected prefix. This is the union criterion from Chapter 4. An occurrence in two arms is charged twice in Equation 5.5, although a pooled reader may later deduplicate it.

The learned policy is evaluated on BM25 rankings generated by a sequential execution. Each subquestion is answered using its first five retrieved passages; that predicted answer resolves references in later subquestions. The run retains 200 ranks per arm. Allocation replays these saved lists. Changing B therefore preserves the intermediate answers and downstream queries from the five-passages-per-hop trace-generation policy. The experiment measures context selection conditional on that policy, rather than a fresh budget-dependent execution of the reasoning chain.

The balanced comparator assigns $\lfloor B/A \rfloor$ slots to every arm and distributes the remainder to the earliest arms. The learned rule also starts with one slot per arm. Both comparisons use the same total budget when $B \geq A$. The retained traces contain at most seven arms, so the budgets 8, 16 and 32 reported below satisfy that condition. Zero-depth arms are allowed in the gold-informed oracle from Chapter 4; it is a reference with a broader feasible set than these floor-constrained policies.

5.3.2 Predicting Per-Arm Depth Targets from Recorded Rankings

Ownership and allocation use assignments for different purposes. Chapter 4's classifier estimates which subquestion needs a passage. Here, training uses the arm that retrieves a passage earliest to define a depth target. These allocation labels come from the recorded

ranks, not the ownership classifier. The learned policy then predicts how many passages to retain from each arm.

Training assigns each supporting passage to the arm that ranks it earliest, breaking ties by arm order. An arm's target depth d_a is the largest rank among its assigned passages, with target one for an empty assignment. If any required passage is absent from every retained list, all arm targets for that question are censored beyond the maximum depth. This convention couples the training labels across arms.

For each threshold in

$$\mathcal{K} = \{1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 48, 64, 96, 128, 200\},$$

a binary gradient-boosted classifier predicts

$$\widehat{S}_a(k) = \widehat{\Pr}(d_a \leq k \mid x_a). \tag{5.6}$$

The implementation fits separate threshold classifiers and then applies a running maximum over their predictions. This creates non-decreasing cumulative curves. It is an upward monotone envelope, with probability calibration remaining a separate property to assess.

Features describe arm text, position, arm count and the first ten BM25 scores. The score features include their spread, concentration, selected gaps and decay. They are computed from the saved rankings before prefix selection. Their acquisition is outside the passage-slot accounting in Equation 5.5. The reported budget therefore measures selected context, rather than total retrieval work, latency or model cost. On sequential traces, later resolved query text also depends on the earlier five-passage reader calls.

5.3.3 Segment and Unit-Increment Allocation

We compare two ways to distribute the budget using the predicted depth curves. The segment policy adds blocks of passage slots, while the unit-increment policy adds one slot at a time. Both have coverage results; the completed final-reader experiment evaluates the segment policy.

The allocation score is the separable surrogate

$$\max_{k_a \geq 1,\ \sum_a k_a \leq B} \sum_{a=1}^{A} \log \widehat{S}_a(k_a). \tag{5.7}$$

It treats satisfaction of the assigned arm requirements as conditionally independent. Union coverage remains the evaluation criterion: evidence retrieved by another arm can satisfy the question even when the assigned arm misses it.

The segment policy forms an upper concave envelope of each log curve. It takes affordable whole envelope segments in descending gain-per-slot order, then distributes unused slots to arms with the largest remaining probability mass. This fills the budget, subject to the retained-depth cap. It is a heuristic for Equation 5.7, including when the original log curves are concave. For example, on depths $\{1, 2, 3\}$, log curves $(-4, -3.5, -3)$ and $(-3, -2, -1)$ with $B = 3$ produce allocation $(2, 1)$ and score -6.5, while $(1, 2)$ achieves -6. The remainder rule chooses the lower-confidence arm rather than the larger next marginal gain.

The unit-increment policy instead allocates one slot at a time. Let h_a be the piecewise-linear upper concave envelope of $\{(k, \log \widehat{S}_a(k)) : k \in \mathcal{K}\}$. Starting from $k_a = 1$ for every arm, select an uncapped arm maximising

$$\Delta_a(k_a) = h_a(k_a + 1) - h_a(k_a) \tag{5.8}$$

and increment its depth. The retained cap is 200. Once every marginal gain is zero, additional slots can be assigned to any uncapped arm without changing the objective. With the non-decreasing predicted curves, this rule solves

$$\max_{k_a \in \{1,\ldots,200\},\ \sum_a k_a \le B} \sum_{a=1}^{A} h_a(k_a), \qquad B \ge A. \tag{5.9}$$

Concavity makes each arm's successive unit gains non-increasing. Every allocation corresponds to taking a prefix of those gains from each arm. Selecting the greatest available gain at each step therefore selects the greatest remaining gain overall while preserving the prefix constraint. An exchange of any smaller selected gain with a larger available gain cannot reduce the objective. Repeating this exchange gives the greedy allocation and establishes its optimality for Equation 5.9.

This exactness concerns the interpolated envelope. The envelope can lie above the original log curve even at a measured depth. For example, on depths $\{1, 2, 3\}$, curves $(-4, -3.9, -1)$ and $(-3, -2, -1)$ with $B = 3$ give unit allocation $(2, 1)$. Its original log score is -6.9, whereas $(1, 2)$ achieves -6. The interpolated and original objectives therefore remain distinct, and both differ from complete union coverage.

An exact finite-grid alternative is available through dynamic programming. Let $V_a(b)$ be the best surrogate score for the first a arms with at most b slots. Then

$$V_a(b) = \max_{k \in \mathcal{K},\ k \le b} \left\{ V_{a-1}(b - k) + \log \widehat{S}_a(k) \right\}, \tag{5.10}$$

with $V_0(b) = 0$ for $b \ge 0$ and infeasible states equal to $-\infty$. This costs $O(AB|\mathcal{K}|)$ and optimises the declared discrete surrogate. The coverage measurements below compare the segment and unit-increment policies; dynamic programming remains an alternative for the original grid objective. Surrogate optimisation and the gold-informed union oracle solve different allocation problems.

5.3.4 Evaluation Population and Cross-Validation

The saved sequential evaluations retain 824 MuSiQue, 977 2Wiki and 996 HotpotQA questions from the corresponding thousand-question panels. The missing trace counts are 176, 23 and 4, respectively. Results condition on the available traces, whose inclusion can depend on successful generation and execution. Chapter 4 records this population and the fixed-trace baselines.

Five-fold cross-validation splits by question. Every arm of a held-out question is predicted by classifiers trained on the other folds, and feature standardisation is fitted within each training fold. These development-panel estimates evaluate allocation on questions excluded from the corresponding classifier fit. Source overlap and model-selection scope are discussed in Section 5.5.2.

Reference-decomposition results use benchmark subquestions and form a separate regime. Tables 5.6 and 5.7 use only generated sequential traces with predicted bindings. The generated-plan assignment labels are obtained from earliest passage ranks rather than an assumed alignment between arm indices and benchmark hop indices.

5.3.5 Evidence Coverage and Required Budgets

At the eight-slot budget, both learned policies increase the measured complete-evidence coverage point estimate relative to balanced allocation on all three benchmarks, although the magnitude varies substantially across datasets. Table 5.6 compares their results with balanced allocation across the tested budgets.

Table 5.6: Allocation on fixed BM25 traces. Segment denotes the segment heuristic and Unit the unit-increment optimiser of the concave interpolant. The budget counts selected slots, excluding trace generation and shallow-score acquisition. Differences are percentage points relative to balanced allocation.

Dataset	Budget	Balanced	Segment	Unit	Δ_{Unit} (pp)
MuSiQue	8	0.413	0.445	0.460	+4.7
	16	0.528	0.541	0.546	+1.8
	32	0.607	0.604	0.609	+0.2
2Wiki	8	0.688	0.746	0.751	+6.3
	16	0.864	0.884	0.892	+2.8
	32	0.926	0.932	0.932	+0.6
HotpotQA	8	0.762	0.763	0.770	+0.8
	16	0.859	0.858	0.856	-0.3
	32	0.907	0.904	0.904	-0.3

At $B = 8$, unit-increment allocation raises complete-evidence coverage from 0.688 to 0.751 on 2Wiki and from 0.413 to 0.460 on MuSiQue. The corresponding segment-policy coverages are 0.746 and 0.445. On HotpotQA, unit-increment coverage is 0.770 against the balanced 0.762 at eight slots, and falls below balanced coverage at sixteen and thirty-two slots. Exact optimisation of the envelope thus has budget-dependent effects on measured union coverage. A position-weighted allocation reaches 0.437 on MuSiQue at $B = 8$, compared with the learned segment policy's 0.445; its corresponding 2Wiki coverage is 0.625. These comparators help distinguish the benefit of predicting depth from that of using a simple uneven split.

Table 5.7: Smallest tested passage-slot budget reaching a coverage target on the fixed generated sequential traces, using the segment policy. The oracle uses gold passage ranks and permits zero-depth arms. Ratios compare discrete operating points.

Dataset	Target	Balanced	Segment	Oracle	Balanced/segment
2Wiki	0.70	10	8	6	1.25
	0.90	24	20	10	1.20
MuSiQue	0.50	16	12	8	1.33
	0.60	32	32	16	1.00
	0.70	128	96	48	1.33
HotpotQA	0.75	8	8	6	1.00
	0.90	32	32	20	1.00

Table 5.7 gives a second view of the benefit: the first tested budget reaching a target coverage. MuSiQue reaches 50% coverage at 12 slots under segment allocation and 16 under balanced allocation, a selected-slot ratio of 1.33. These comparisons use the tested budget grid and exclude the common trace-generation cost. The next experiment measures answer quality from the selected contexts.

5.3.6 Paired Final-Answer Evaluation

To test whether improved context selection also benefits answering, we compare the segment allocator with balanced allocation using a common final reader. The completed experiment takes the first 700 retained MuSiQue traces and refits the segment-policy allocator using five-fold question-level cross-validation within those 700 cases. Its coverage values therefore refer to a different subset and fitted models from the 824-question frontier. Each budget–policy pair is evaluated on every selected question, giving 4 200 completed final-reader calls. These outcomes were generated before the unit-increment revision and measure the segment policy throughout.

Both policies deduplicate and interleave their selected prefixes. The same GPT-4o-mini reader, requested through the API's model alias at temperature zero, answers the original question from that pooled context. This differs from Chapter 4's sequential execution, which returns the last arm's answer. Here the intermediate trace is preserved, and a separate final reader tests the contexts selected by each allocation policy. Table 5.8 reports the paired outcomes.

Table 5.8: Pooled final-reader evaluation on 700 retained MuSiQue traces. U denotes balanced allocation and L the learned segment heuristic. All 4,200 question–budget–policy evaluations completed. The last column gives L minus U and its descriptive pointwise paired 95% F1 interval; the intervals at budgets 16 and 32 contain zero. The unit-increment allocator is a separate policy.

Budget	Cov. U	Cov. L	EM U	EM L	F1 U	F1 L	ΔF1 [95% CI]
8	0.413	0.449	0.344	0.354	0.424	0.445	$+0.021$ $[0.004, 0.039]$
16	0.523	0.547	0.360	0.377	0.454	0.464	$+0.010$ $[-0.008, 0.027]$
32	0.603	0.600	0.367	0.359	0.472	0.458	-0.014 $[-0.031, 0.001]$

At $B = 8$, segment allocation raises F1 from 0.4244 to 0.4455 and exact match from 0.3443 to 0.3543. The paired F1 difference is 0.0211, with a descriptive 95% interval of $[0.0038, 0.0385]$ from 10 000 question-level bootstrap resamples. The corresponding F1 differences at $B = 16$ and $B = 32$ are 0.0098 and -0.0144, with intervals $[-0.0076, 0.0270]$ and $[-0.0309, 0.0014]$. At eight slots, the segment policy produces a 2.1-point higher F1 on this development panel, with a pointwise paired 95% interval of [0.0038, 0.0385]. This interval is not adjusted for comparisons across multiple budgets; results at 16 and 32 slots are smaller and their intervals include zero. The pointwise intervals at the larger budgets include zero; Section 5.5.2 discusses the scope of the comparison.

Every allocation spends its declared prefix budget. Deduplication leaves mean context sizes of 7.37 versus 7.39 passages at $B = 8$, and 30.04 versus 30.31 at $B = 32$. Thus the matched quantity is selected slots, with unique passages and token lengths free to differ. The observed answer gain at the smaller budget concerns this pooled-context intervention. Changing an intermediate reading budget requires regeneration of its dependent queries to test a sequential execution policy.

An exploratory study also tests hidden-state features for predicting assigned depth. Its results and methodological qualifications are retained in Appendix A.1; those features are not used by the allocation policies evaluated above.

5.4 Evidence-Aware Feedback: Design and Diagnostic Validation

The third contribution turns Chapter 4's evaluation into structured execution feedback. We implement a representation that organises evidence availability, estimated ownership, and intermediate-answer information so that a prompt optimiser can inspect where an execution needs improvement. The interface supports comparisons with answer-only and raw evidence feedback. This section presents its design and completed ownership checks, followed by the proposed evaluation of its effect on prompt search.

5.4.1 Separating Reward from Diagnostic Information

A scalar reward ranks candidate instructions. A diagnostic record identifies features of their executions that the optimizer can use when proposing a revision. Keeping these roles separate allows the same final-answer reward to be paired with different feedback representations.

For a question q, the implemented record contains the generated plan, resolved queries, intermediate answers, retrieved evidence and final-answer score. The gold passages and reference decomposition are available to the optimisation-side feedback service. During execution, the decomposer and reader use the question, fixed demonstrations, retrieved passages and their own predicted bindings. Gold intermediate answers enter feedback after the execution rather than resolving its dependencies.

The implemented feedback builder supports three views:

1. **Answer feedback.** The question, generated plan, final answer, reference answers, exact match and F1.

2. **Raw evidence feedback.** Answer feedback plus reference obligations and their passage text, executed queries, intermediate predictions and passage ranks. Available cell votes from the frozen ownership ensemble are included as measurements.

3. **Structured evidence feedback.** The same raw evidence payload plus explicit dependency edges, per-obligation retrieval status, estimated ownership status and restricted binding checks.

The raw and structured views share identical underlying evidence text and ownership measurements for a given trace. Their shared payload is hashed after common clipping, and both obey the same overall reflection input cap. Structured feedback adds a derived organisation of that information. Its comparison with raw feedback therefore tests the value of the representation, while the comparison with answer-only feedback also includes additional supervision. Actual reflection tokens remain a reported cost because equal caps do not imply equal realised input lengths.

5.4.2 Observed Events and Estimated Assignments

Table 5.9 distinguishes deterministic observations from classifier-dependent interpretations. The distinction prevents a failed answer from being assigned a single cause on the basis of a score alone.

Table 5.9: Diagnostic fields supplied by the structured feedback interface.

Field	Measurement	Scope
Dependency structure	Plan references and interface validity	A property of the generated plan
Reader evidence gap	Required passage absent from every five-passage reader context	An observation of this execution
Retained-list gap	Required passage absent from all saved rankings	Censored at the retained depth
Unassigned obligation	An obligation with zero estimated owners	An estimated assignment gap
Fold disagreement	Votes differ for an arm–evidence pair	Estimator uncertainty
Binding comparison	Intermediate answer compared with an aligned reference	Available only for a uniquely matched template and its dependencies

The binding check compares intermediate predictions with reference answers when normalised subquestion templates and their dependencies align uniquely. Paraphrased, merged, or otherwise ambiguous arms remain unmatched. The record therefore makes explicit both the available comparison and the cases where alignment cannot support one.

Ownership estimates retain the five model votes and the identity of the frozen estimator. A predicted unowned passage is described as a suspected unassigned obligation. When a passage is available in a different arm's context, the record distinguishes global availability from the evidence available to its estimated owner. This is the distinction that the union and masked depths formalised in Chapter 4.

5.4.3 Ownership Diagnostics under Controlled Plan Changes

The completed stress suite tests the ownership diagnostics on known plan transformations. Twelve questions from the frozen optimisation split are selected deterministically before classifier evaluation, with four at each of two, three and four hops and twelve distinct source components. They yield 52 cases: 12 reference plans, 12 relevant-arm duplications, 12 irrelevant-arm insertions, 12 sink-arm omissions and 4 independent-arm fusions.

Expected ownership masks are derived from transformation provenance. Duplication adds an owner for the same supporting passage. An irrelevant arithmetic query adds an arm with no obligation from the reference task. Omitting an unreferenced sink creates a missing obligation only when its supporting passage is unique among the remaining arms. Shared evidence is tracked separately so that deleting a hop is not automatically equated with deleting its passage coverage.

The four fusions combine independent reference arms whose distinct answers are subsequently used downstream. The scalar reference grammar provides no selector for multiple outputs from one fused arm. These four cases therefore use explicitly gold-resolved downstream inputs and test ownership on synthetic plans. They are excluded from claims about executable fused policies. The remaining transformations preserve valid backward references.

The frozen five-model ensemble was evaluated on all 52 cases using the reconstructed feature pipeline. Table 5.10 compares the predicted mask with the complete provenance-derived mask for each case. These transformation-suite measurements are separate from the historical classifier evaluation in Chapter 4.

Table 5.10: Ownership predictions on the optimisation-only stress suite. Exact agreement requires every cell of a case's ownership mask to match the transformation labels. Variants share twelve base questions.

Transformation	Cases	Exact masks
Reference plan	12	12
Duplicate relevant arm	12	12
Insert irrelevant arm	12	7
Omit sink arm	12	6
Fuse independent arms, gold-resolved	4	2

The ensemble preserves the expected masks for all twelve reference plans and all twelve relevant-arm duplications. It also recovers the full mask in seven irrelevant-insertion cases, six omission cases, and two fusion cases. The stress suite identifies transformations for which the estimator is stable and others for which ownership predictions become unreliable. These results motivate retaining model votes and expressing unassigned obligations as suspected rather than definitive in the feedback interface. These results inform the feedback design: ownership estimates and model disagreement remain visible, and an unassigned obligation is described as suspected.

5.4.4 Reflective Search with a Fixed Execution Interface

GEPA supplies a reflective prompt-search procedure that uses execution feedback and candidate performance across examples [34]. The extension uses the pinned GEPA implementation to optimise the single decomposer instruction. The same adapter evaluates answer, raw and structured feedback conditions. The proposed contribution is the evidence-feedback representation within this common search procedure.

The planned comparison uses final-answer F1 as its scalar reward in every condition. The original masked-depth score, complete-evidence coverage and resource use are retained as diagnostics. Candidate selection uses the instance-level Pareto strategy, with reflection batches of eight optimisation examples and strict improvement acceptance. Examples are sampled by shuffled epochs, and their exposure counts are logged.

The search optimises a single text component, the decomposer instruction. It uses reflective mutation with the component-exchange merge operator disabled. This keeps the intervention focused on prompt revision under the common execution interface.

The execution contract permits one to six arms and strictly backward references. Invalid plans trigger a shared direct-question fallback, which remains in the quality and cost denominators. The reader uses five passages per arm, and the same model settings, demonstrations and passage corpus are used across fresh search conditions. This matches the maximum execution allowance while leaving actual arm count and tokens free to differ.

The runner uses GPT-4o-mini, snapshot `gpt-4o-mini-2024-07-18`. Reflection uses GPT-4o, snapshot `gpt-4o-2024-08-06`. The runner checks both returned model identifiers. The declared task seed and first search seed are one, with temperature zero for execution and one for reflection. Provider identity and generation settings enter the cache

and run fingerprints. These settings remain separate from the historical Llama-based pilot.

5.4.5 Proposed Evaluation of Feedback Quality

The proposed comparison uses final-answer F1 to select instructions and varies the feedback supplied to a common search procedure. Answer-only feedback provides the basic comparator. Raw and structured evidence feedback share the same underlying information, so their comparison tests the value of organising that information into explicit diagnostics.

The frozen design separates 96 optimisation questions, 48 selection questions, and 200 confirmation questions, with no shared single-hop source IDs between these splits. The questions are new compositions whose supporting passages are present in the fixed corpus; historical subproblem reuse remains. Each search receives the same ceilings of 512 example-level executions and 24 candidate proposals. Final policies are compared on paired answer quality, evidence coverage, and actual search and inference costs.

Appendix A.2 preserves the complete data-selection procedure, comparators, allowances, and analysis plan. It also specifies a future comparison of prompt and allocation effects. These protocols define how to assess the extension; they are distinct from the completed ownership checks and the prompt and allocation results reported above.

5.5 Discussion and Implications

5.5.1 Contributions and Practical Implications

The chapter develops two measured applications of Chapter 4's framework. The OPRO pipeline demonstrates how ownership-masked retrieval evaluation can drive automated instruction search. On the shared development panel, the search selects a prompt with a modestly higher measured depth score, while finite coverage and exact match remain unchanged and aggregate F1 is essentially unchanged. The candidate comparison also shows that retrieval depth and final-answer F1 can favour different instructions, making objective choice an explicit design decision.

The learned allocator turns the oracle opportunity into decisions based on query features and retrieval scores. At eight slots, unit-increment allocation improves complete-evidence coverage on all three benchmarks. The separate segment-policy reader experiment provides an answer-level result on 700 MuSiQue questions: F1 increases by 2.1 points at that budget. These findings support learned passage selection as a promising intervention in the tight-context regime evaluated here, while the mixed larger-budget results show that its advantage is not uniform across operating points.

The structured-feedback interface connects the third foundation to prompt revision. It preserves the evidence available to each arm, the predicted ownership assignments, and intermediate-answer comparisons in a common record. Its controlled checks establish how the ownership diagnostics respond to specific plan changes. The information-matched raw view and frozen evaluation protocol make the effect of organising that information testable within a common search procedure.

5.5.2 Scope and Limitations

The prompt-search results come from one initial instruction and four rounds on the same 100 questions used for feedback and selection. The 0.90-point score gain is modest, its descriptive bootstrap interval includes zero, and exact match remains 44%. The selected prompt also increases the mean passage allowance from 15.00 to 15.65 slots. Independent questions and additional search runs are needed to assess generalisation and variability. The retrospectively identified F1-leading candidate is a diagnostic comparison.

Allocation cross-validation separates questions, but shared atomic subproblems and supporting documents can recur across folds. Method choices were developed on these panels. The paired reader intervals are pointwise, without adjustment across budgets or resampling shared source components. The larger-budget reader results are mixed: the segment policy gains 1.0 F1 point at sixteen slots and loses 1.4 points at thirty-two, with both intervals including zero. The unit-increment policy has coverage measurements but no corresponding final-reader evaluation.

Both allocation policies use rankings produced by five-passage sequential reader calls. Selected slots exclude trace generation and feature acquisition, and equal slot budgets can yield different deduplicated contexts and token counts. The unit-increment rule is exact for its concave-envelope objective, which differs from the predicted grid objective and gold-evidence union coverage. A sequential budget policy would need to regenerate intermediate answers and dependent queries using information available at each step.

The ownership stress suite contains 52 variants of twelve base questions; its four fused cases use gold-resolved downstream inputs. Broader validation requires independently labelled generated plans. Intermediate-answer mismatches provide diagnostic evidence, while causal attribution requires controlled re-execution. The structured-feedback search comparison remains prospective. Its frozen questions are new compositions within a fixed, corpus-covered population with historical subproblem reuse, as documented in Appendix A.2.

Conclusion

This chapter put the evidence-aware framework into use through automated prompt search and learned passage allocation.The OPRO pilot connected the ownership-masked score to instruction selection and selected a candidate with a 0.90-point higher depth score on the shared development panel, while finite coverage and exact match remained unchanged. Because search and selection used the same questions, this result demonstrates the optimisation loop rather than out-of-sample prompt improvement. The comparison of candidate instructions further showed how the choice of objective shapes the selected decomposition.

Learned allocation captured part of the opportunity identified by Chapter 4's oracle: at eight slots, unit-increment coverage increased by 6.3 percentage points on 2Wiki and 4.7 on MuSiQue. A separate segment-policy experiment on 700 retained MuSiQue questions produced a 2.1-point higher final-answer F1 at the eight-slot budget; its advantage diminished at larger budgets. Finally, the implemented feedback representation makes evidence responsibility and execution events available for prompt revision. Its diagnostic behaviour has been stress-tested, but whether this structured feedback improves prompt search relative to answer-only or raw evidence feedback remains to be evaluated under the prospective protocol.

Conclusion and outlook

This internship project aimed to design

Appendix A
Supplementary Analyses and Evaluation Protocol

This appendix preserves the exploratory depth-prediction study and the detailed evaluation protocol supporting Chapter 5. The first reports completed probe measurements; the second describes the proposed feedback comparison. The main chapter presents the completed prompt-search, allocation, and ownership-check results.

A.1 Exploratory Hidden-State Prediction of Assigned Depth

Chapter 3 studies representations of compositional hop count. A separate probe examines their association with the assigned depth target d_a defined in Section 5.3. The panel contains 2 648 reference arms from 1 000 MuSiQue questions with BM25 rankings. Supporting passages are assigned to the earliest-ranking arm; 31 arm targets belong to questions whose required evidence is unreachable within the retained lists. The binary targets are $d_a \leq k$ for $k \in \{2, 4, 8, 16, 32\}$.

The frozen models encode the reference subquestion templates as raw text, without a chat template. Right-padding-aware last-token pooling supplies the features at nine sampled layers per model. Of the 2 648 templates, 1 517 contain unresolved `#N` references and differ from the gold-resolved queries used for retrieval. The representations therefore describe the templates rather than the executed queries.

Table A.1: Exploratory depth prediction on the fixed reference panel. Mean AUROC averages the five depth thresholds. Each model's layer is selected using the same cross-validation results reported here.

Features	Selected layer	Mean AUROC
Qwen2.5-7B-Instruct	14	0.8397
Llama-3.1-8B-Instruct	12	0.8454
Mistral-7B-Instruct-v0.3	16	0.8442
Surface features and top-ten BM25 scores	n/a	0.7639

Five-fold cross-validation keeps all arms of a question together. Feature standardisation is fitted within each training fold. Logistic regression uses inverse regularisation strength $C = 0.01$ for hidden states and $C = 1$ for the surface-and-score comparator,

so Table A.1 compares two specified readouts as well as their features. Near-constant hidden-state dimensions are filtered using the full panel before splitting.

The split retains substantial source overlap: 1 557 test-arm instances have an identical template in their training fold, and 755 test questions share a source subproblem with training. Layer selection on the reported folds adds selection optimism. All selected-layer activations are finite; the Qwen final-layer arrays contain nonfinite activations for 2 237 arms, which the probe code replaces with zero. That layer is unsuitable for interpreting a layer-wise performance decline.

These measurements motivate a component-separated evaluation of template representations with nested layer selection and matched readout tuning. Their use in an allocator requires a further end-to-end comparison of evidence coverage, answer quality and representation-extraction cost.

A.2 Independent Evaluation Design

A.2.1 Frozen Data Roles and Corpus Scope

The proposed feedback comparison separates questions used for reflection, questions used for candidate selection and questions reserved for final confirmation. The source is the official answerable MuSiQue development release [4]. Excluding all composite question IDs in the historical thousand-question panel leaves 1 417 cases. Removing three additional exact normalised question-text matches leaves 1 414.

Of those questions, 363 have every gold supporting passage in the fixed 11 656-passage corpus. The initial matched-backbone study uses this eligible subset. Eligibility is determined before new model outputs, by exact passage identity rather than observed retrieval success. It defines a corpus-covered target population. Broader MuSiQue performance requires a separate evaluation with the additional corpus material included.

Questions sharing any underlying single-hop source ID are grouped before the new splits are assigned. The final manifests contain 96 optimisation, 48 selection and 200 confirmation questions, with zero shared source IDs across these three splits. Table A.2 gives their composition. Split identities and file hashes are frozen before candidate generation.

Table A.2: Frozen question-new splits for the prospective comparison. Components connect questions sharing a source single-hop ID.

Role	Questions	Two-hop	Three-hop	Four-hop	Components
Optimisation	96	23	40	33	29
Selection	48	16	19	13	22
Confirmation	200	73	77	50	65

All selected composite question IDs and texts differ from the historical panel and the ownership classifier's training panel. Historical subproblem reuse remains: 343 of the 344 selected questions share at least one single-hop source with the historical thousand questions. These are new compositions under the documented exposure exclusions, rather than wholly new atomic facts. There is no source-ID overlap with the classifier's training questions. The confirmation set therefore tests transfer to new compositions within the corpus-covered regime.

The reflection service accepts only hash-bound optimisation samples. The selection set supplies scores for ranking candidates and remains adaptive development data. Confirmation question contents and labels stay outside the search interface until the final instructions and analysis choices have been fixed.

A.2.2 Comparators and Search Allowances

The fresh study includes the unchanged initial instruction, nonadaptive instruction sampling, OPRO-style answer-feedback search and three GEPA feedback conditions. Nonadaptive proposals are generated from the task and initial instruction before their evaluation results are available. The OPRO control uses score histories and answer errors. The three GEPA conditions share the optimizer, reward and execution interface while varying the feedback view.

Each fresh search is assigned a ceiling of 512 example-level rollout evaluations and 24 candidate proposals. A rollout is an executed candidate–question pair. Parent comparisons, proposed-candidate comparisons and validation executions count toward that ceiling. The different methods allocate their allowance differently, so search progress is measured against cumulative evaluations and actual cost.

The nonadaptive schedule screens 24 instructions on a common set of 12 optimisation questions, then evaluates its four leaders on the 48 selection questions. This uses

$$24 \times 12 + 4 \times 48 = 480 \tag{A.1}$$

candidate–question evaluations. The OPRO control uses four rounds of six proposals, with a declared twelve-question screen per round and selection evaluation for each round's nominee. The reflective search uses its allowance for parent and child batches and complete selection evaluations. The number of completed mutations is reported explicitly rather than inferred from the evaluation ceiling.

Within a search condition, the winner is selected by mean selection F1. Ties are resolved by lower mean execution tokens and then instruction hash. The fixed initial instruction and the historical OPRO winner are retained as identified reference policies. The fresh pipeline may use a different API model configuration from the historical Llama pilot; comparisons are made within a frozen fresh backbone, and historical scores remain in their original regime.

The principal contrast is structured versus answer-only feedback. Structured versus raw feedback is required to isolate the effect of diagnostic organisation from additional supervision. A second predeclared search seed provides a replication of search variability when resources permit. Both seeds' trajectories must then be retained.

A.2.3 Outcome Analysis and Cost Accounting

After selection, each final policy is evaluated on the same locked confirmation questions. The primary outcome is the question-weighted mean paired F1 difference. Exact match, complete evidence in the reader contexts, union depth, masked depth, arm count and model tokens are secondary measurements. All questions, including fallback executions, contribute to the aggregate results.

Paired bootstrap resampling preserves each question's outcomes across policies. A complementary component-level bootstrap keeps questions sharing a source component

together. The confirmation panel contains 65 such components, with at most 20 questions in one component. Question and component resampling describe sensitivity under different sampling units; neither supplies a distribution-free generalisation guarantee.

Search cost and inference cost are recorded separately. Search cost includes instruction proposals, reflection and candidate evaluation. Inference cost includes decomposition, retrieval-associated model calls and every reader step for the selected policy. Cached deterministic evaluations count toward the logical search allowance while their actual charges are recorded separately. A cache hit is reuse of a result, not an independent execution.

The passage-allocation replay in Chapter 4 provides a complementary cost diagnostic. On recorded lists it compares balanced total-slot allocation with the smallest oracle prefix allocation covering the gold passages. The oracle has access to gold identities and frozen rankings. It measures available allocation headroom; it is not a learned execution policy. In the sequential pipeline, changing an early reader allowance may change its answer and every downstream query. Such a policy requires fresh execution before claiming an answer-quality or runtime gain.

A.2.4 Separating Prompt and Allocation Effects

The completed studies intervene at different points: the OPRO pilot changes the decomposition instruction, while Section 5.3 changes prefixes selected from fixed rankings. A prospective factorial comparison can test their combination. Freeze an initial prompt p_0, a search-selected prompt p_1, balanced allocation g_0 and learned allocation g_1 before final evaluation. Evaluate all four pairs (p_i, g_j) on the same questions and reader budget. If Y_{ij} denotes mean final-answer F1, the interaction is

$$I = (Y_{11} - Y_{10}) - (Y_{01} - Y_{00}). \tag{A.2}$$

It asks whether the allocation benefit changes under the new prompt. Each prompt generates its own traces under the same five-passages-per-hop policy; both allocators then act on those traces. This comparison concerns pooled context, with trace-generation cost reported separately. Its outcomes remain prospective.

A sequential extension must choose each arm's budget using information available at that step and reserve resources for later arms. It then recomputes intermediate answers and downstream retrieval. The offline allocator uses resolved later queries and their score profiles, which would give an online policy future information. Step-local allocation and re-execution are therefore part of that extension.

Bibliography

[1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 9459–9474, 2020.

[2] Z. Yang, P. Qi, S. Zhang, Y. Bengio, W. W. Cohen, R. Salakhutdinov, and C. D. Manning, "HotpotQA: A dataset for diverse, explainable multi-hop question answering," in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pp. 2369–2380, 2018.

[3] X. Ho, A.-K. D. Nguyen, S. Sugawara, and A. Aizawa, "Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps," in *Proceedings of the 28th International Conference on Computational Linguistics*, pp. 6609–6625, 2020.

[4] H. Trivedi, N. Balasubramanian, T. Khot, and A. Sabharwal, "MuSiQue: Multihop questions via single-hop question composition," *Transactions of the Association for Computational Linguistics (TACL)*, vol. 10, pp. 539–554, 2022.

[5] O. Abdelhedi, O. R. Heidari, S. Chaari, X. Chen Zhu, M. Hamed, D. Azzam, and Y. Yaakoubi, "Probing compositional question difficulty in language model representations," in *IEEE AIDIST 2026 Conference*, 2026. Accepted, to appear.

[6] G. Salton and C. Buckley, "Term-weighting approaches in automatic text retrieval," *Information Processing & Management*, vol. 24, no. 5, pp. 513–523, 1988.

[7] S. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.

[8] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W.-t. Yih, "Dense passage retrieval for open-domain question answering," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 6769–6781, 2020.

[9] O. Khattab and M. Zaharia, "ColBERT: Efficient and effective passage search via contextualized late interaction over BERT," in *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 39–48, 2020.

[10] W. Xiong, X. L. Li, S. Iyer, J. Du, P. Lewis, W. Y. Wang, Y. Mehdad, W.-t. Yih, S. Riedel, D. Kiela, and B. Oğuz, "Answering complex open-domain questions with

multi-hop dense retrieval," in *International Conference on Learning Representations (ICLR)*, 2021.

[11] O. Khattab, C. Potts, and M. Zaharia, "Baleen: Robust multi-hop reasoning at scale via condensed retrieval," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021.

[12] H. Trivedi, N. Balasubramanian, T. Khot, and A. Sabharwal, "Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions," in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 10014–10037, 2023.

[13] B. J. Gutiérrez, Y. Shu, Y. Gu, M. Yasunaga, and Y. Su, "HippoRAG: Neurobiologically inspired long-term memory for large language models," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[14] B. J. Gutiérrez, Y. Shu, W. Qi, S. Zhou, and Y. Su, "From RAG to memory: Non-parametric continual learning for large language models," in *Proceedings of the 42nd International Conference on Machine Learning*, vol. 267 of *Proceedings of Machine Learning Research*, pp. 21497–21515, PMLR, 2025.

[15] S. Jeong, J. Baek, S. Cho, S. J. Hwang, and J. C. Park, "Adaptive-RAG: Learning to adapt retrieval-augmented large language models through question complexity," in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 7036–7050, 2024.

[16] Z. Jiang, F. Xu, L. Gao, Z. Sun, Q. Liu, J. Dwivedi-Yu, Y. Yang, J. Callan, and G. Neubig, "Active retrieval augmented generation," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 7969–7992, 2023.

[17] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-RAG: Learning to retrieve, generate, and critique through self-reflection," in *International Conference on Learning Representations (ICLR)*, 2024.

[18] G. Alain and Y. Bengio, "Understanding intermediate layers using linear classifier probes," *arXiv preprint arXiv:1610.01644*, 2016.

[19] K. Park, Y. J. Choe, and V. Veitch, "The linear representation hypothesis and the geometry of large language models," in *Proceedings of the 41st International Conference on Machine Learning*, vol. 235 of *Proceedings of Machine Learning Research*, pp. 39643–39666, PMLR, 2024.

[20] J. Hewitt and P. Liang, "Designing and interpreting probes with control tasks," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pp. 2733–2743, Association for Computational Linguistics, 2019.

[21] S. Kadavath, T. Conerly, A. Askell, T. Henighan, D. Drain, E. Perez, N. Schiefer, Z. Hatfield-Dodds, N. DasSarma, E. Tran-Johnson, S. Johnston, S. El-Showk, A. Jones, N. Elhage, T. Hume, A. Chen, Y. Bai, S. Bowman, S. Fort, D. Ganguli,

D. Hernandez, J. Jacobson, J. Kernion, S. Kravec, L. Lovitt, K. Ndousse, C. Olsson, S. Ringer, D. Amodei, T. Brown, J. Clark, N. Joseph, B. Mann, S. McCandlish, C. Olah, and J. Kaplan, "Language models (mostly) know what they know," *arXiv preprint arXiv:2207.05221*, 2022.

[22] C. Burns, H. Ye, D. Klein, and J. Steinhardt, "Discovering latent knowledge in language models without supervision," in *The Eleventh International Conference on Learning Representations*, 2023.

[23] S. Marks and M. Tegmark, "The geometry of truth: Emergent linear structure in large language model representations of true/false datasets," in *First Conference on Language Modeling*, 2024.

[24] A. Azaria and T. Mitchell, "The internal state of an LLM knows when it's lying," in *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 967–976, Association for Computational Linguistics, 2023.

[25] I. V. Moreno Cencerrado, A. Padrés Masdemont, A. Gonzalvez Hawthorne, D. D. Africa, and L. Pacchiardi, "No answer needed: Predicting LLM answer accuracy from question-only linear probes," *arXiv preprint arXiv:2509.10625*, 2025.

[26] S. Yang, E. Gribovskaya, N. Kassner, M. Geva, and S. Riedel, "Do large language models latently perform multi-hop reasoning?," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 10210–10229, 2024.

[27] O. Press, M. Zhang, S. Min, L. Schmidt, N. A. Smith, and M. Lewis, "Measuring and narrowing the compositionality gap in language models," in *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 5687–5711, 2023.

[28] S. Min, V. Zhong, L. Zettlemoyer, and H. Hajishirzi, "Multi-hop reading comprehension through question decomposition and rescoring," in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 6097–6109, 2019.

[29] E. Perez, P. Lewis, W.-t. Yih, K. Cho, and D. Kiela, "Unsupervised question decomposition for question answering," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 8864–8880, 2020.

[30] D. Dua, S. Gupta, S. Singh, and M. Gardner, "Successive prompting for decomposing complex questions," in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 1251–1265, 2022.

[31] D. Zhou, N. Schärli, L. Hou, J. Wei, N. Scales, X. Wang, D. Schuurmans, C. Cui, O. Bousquet, Q. V. Le, and E. H. Chi, "Least-to-most prompting enables complex reasoning in large language models," in *International Conference on Learning Representations (ICLR)*, 2023.

[32] T. Khot, H. Trivedi, M. Finlayson, Y. Fu, K. Richardson, P. Clark, and A. Sabharwal, "Decomposed prompting: A modular approach for solving complex tasks," in *International Conference on Learning Representations (ICLR)*, 2023.

[33] C. Yang, X. Wang, Y. Lu, H. Liu, Q. V. Le, D. Zhou, and X. Chen, "Large language models as optimizers," in *International Conference on Learning Representations*, 2024.

[34] L. A. Agrawal, S. Tan, D. Soylu, N. Ziems, R. Khare, K. Opsahl-Ong, A. Singhvi, H. Shandilya, M. J. Ryan, M. Jiang, C. Potts, K. Sen, A. G. Dimakis, I. Stoica, D. Klein, M. Zaharia, and O. Khattab, "GEPA: Reflective prompt evolution can outperform reinforcement learning," in *International Conference on Learning Representations*, 2026.

Netography

[N1] Concordia University, “The birth of Concordia: Loyola College and Sir George Williams University merged on 24 August 1974.” https://www.concordia.ca/cunews/main/stories/2015/08/24/archives-loyola-and-sir-george-williams-merge.html. Accessed: 2026-08-13.

[N2] Concordia University, “Sir George Williams University history.” https://www.concordia.ca/offices/archives/stories/sgw.html. Accessed: 2026-08-13.

[N3] Concordia University, “Concordia fast facts: student enrolment 2024–25.” https://www.concordia.ca/about/fast-facts.html. Accessed: 2026-08-13.

[N4] Concordia University, “First engineering faculty in Canada named after a woman: business leader Gina Cody makes historic $15 million donation.” https://www.concordia.ca/news/stories/2018/09/24/gina-cody-makes-historic-eponymous-15m-donation-to-have-canadas-first-engineering-faculty-named-after-a-woman.html. Published 24 September 2018. Accessed: 2026-08-13.

[N5] Concordia University, “Gina Cody — notable Concordians.” https://www.concordia.ca/profiles/offices/advancement/notable-concordians/gina-cody.html. Accessed: 2026-08-13.

[N6] Concordia University, “About the school — gina cody school of engineering and computer science.” https://www.concordia.ca/ginacody/about.html. Accessed: 2026-08-13.